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Incentives, penalties, and rural air pollution: Evidence from satellite data

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ABSTRACT

I test the role of economic incentives and command and control in reducing agricultural fires, a major source of air pollution in most rural regions across the world caused by burning crop residues after harvest. To tackle data shortage, I use high-resolution satellite data to construct a fine measure of agricultural fires as well as other geographic characteristics at 1 km × 1 km resolution for China. Using the staggered arrival of biomass power plants, which purchase crop residues as production inputs from nearby areas, as a shock to economic incentives, I find a more than 30% drop in agricultural fires in the vicinity of a plant after its opening relative to areas farther away. Such drop cannot be explained by structural transformation, migration, or enhanced regulation near the plant, and is consistent with an incentive-based explanation. I then examine the effectiveness of a command and control policy that bans agricultural fires within 15 km of airports. Using a spatial regression discontinuity design, I find no evidence that such policy works.

1. Introduction

Burning crop residues after harvest is a long-standing practice across the world, especially in developing countries.[2] It is perceived as a convenient way to prepare land for subsequent cultivation. At the same time, the open burning releases a number of pollutants, which could travel a long distance and generate a wide range of harms to human health (Rangel and Vogl, 2019; Graff Zivin et al., 2020; He et al., 2020a). Despite such negative externalities, how to effectively reduce agricultural fires is largely unexplored.

In practice, there are two widely used instruments to fight pollution: one is based on economic incentives and the other is based on command and control (Oates et al., 1989).[3] For the former, a typical example is to provide subsidies to polluters for pollution abatement; for the latter, a typical example is to punish polluters through regulations and laws. In the context of fighting agricultural fires, which policy could be effective is, however, unclear for two reasons. The first reason is conceptual. Subsidies to farmers could be lower than their opportunity cost of not burning, either due to weak capacity for taxation or corruption (Greenstone and Jack, 2015). Even if subsidies are large enough, the effects are still indeterminate, since farmers could use the subsidies to expand farming, which may lead to more agricultural fires. The effectiveness of command and control also hinges on a number of factors such as the dissemination of the laws among farmers, the capacity to detect and punish burning, and the rule-abiding norms. In sum, disentangling the effects of these policies entails empirical investigation.

The second reason is that empirical tests typically suffer from at least three challenges. First, there exists an inherent selection bias associated with the rollout of a policy. For example, a policy may be launched in certain area following an outburst of agricultural fires. Then the subsequent reduction in agricultural fires could reflect reversion to the mean. Second, many policies have no clear or very aggregated target areas such as cities or provinces, making the results easily contaminated by other concurrent policies or omitted variables. Third, existing measures of fires, which are usually extracted directly from satellite observations, do not differentiate between fires from agricultural activities and from non-agricultural activities. Therefore, studies based on such measures may suffer from measurement errors.

I address these challenges in several steps. First, I use two unique quasi-experiments to control for selection bias. The first is the staggered adoption of biomass power plants in China that purchase crop

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[2] A general description of agricultural fires worldwide can be found in a report by World Bank at https://openknowledge.worldbank.org/handle/10986/29504.

[3] In terms of curbing fires, technological innovation, such as cutting crop residues through specialized harvesters and returning them into the soil, could be a third alternative. However, this would induce a nontrivial cost. Due to the lack of economies of scale in agriculture and farmers' low marginal willingness to pay for clean air, such technology and related machines are not well developed and generalized in many developing countries.

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residues from farmers as inputs, creating an economic incentive for farmers to reduce burning. Crucially, the main purpose of these plants is to expand energy sources and reduce the reliance on oil, instead of curbing agricultural fires. This experiment enables me to adopt a spatial difference-in-differences (DID) strategy to test the effects of economic incentives on agricultural fires. The second is the drawing of no burning zones, which target solely agricultural fires within 15 km of airports in China. The zone boundaries are purely artificially drawn and do not overlap with administrative boundaries or other policy thresholds. This experiment enables me to conduct a spatial regression discontinuity (RD) design to test the effects of command and control on agricultural fires. Second, I rely on micro data at the 1 km × 1 km resolution and include granular fixed effects to control for potential confounding factors such as local economic conditions. Third, I identify agricultural fires by matching satellite fire observations with detailed land use data to tackle the measurement issue. The land use data enable me to differentiate between agricultural fires and other fires.

I justify the validity of the DID strategy in various ways. First, I compare areas very close to a plant with areas that are a bit farther away. Due to spatial adjacency, these areas should be somewhat comparable ex ante, alleviating the concern that areas with and without a plant may be quite different in unobservable aspects. Specifically, I define two treatment groups based on qualitative and quantitative evidence: the most treated group consisting of areas within 5 km of a plant, and the less treated group consisting of areas within 5–10 km of a plant. The control group includes areas within 10–15 km of a plant. Second, I include highly disaggregated fixed effects, such as grid cell, plant × year, and city × year fixed effects, to control for a number of confounding factors. I also include a set of flexible trends specific to each grid cell to rule out the possibility that the DID estimates are just reflecting differential trends across areas. Third, I use an event study specification to show that there is no diverging pre-trends between treatment groups and control groups. Fourth, I conduct two placebo tests using non-agricultural fires and randomly generated plant openings as placebos, respectively. Finally, I directly control for some factors that affect location selection and correct for selection on unobservables by assuming the selection on observables is informative about selection on unobservables.

Using data from 2001 to 2016 on agricultural fires and the openings of 190 biomass power plants, I find a significant reduction in agricultural fires for both the most treated group and the less treated group following the arrival of a plant. In my preferred specification that controls for grid cell fixed effects, plant × year fixed effects, and differential trends across grid cells, the drop in agricultural fires amounts to 57.7% and 30.9% of the sample mean for the most treated group and the less treated group, respectively. The results are robust to correcting for the bias associated with staggered treatment timing. I further verify such decreasing effect with distance using alternative specifications in which I replace the two binary treatment indicators with continuous distance or topography-based distance to a plant. These findings are consistent with the explanation that biomass power plants create economic incentives for farmers to reduce burning, and the economic incentives become weaker when it becomes more costly to transport the crop residues to a plant. I corroborate such explanation using qualitative evidence and also quantitative evidence that shows weaker effects in steeper, more rugged, and more fragmented areas, where it is more difficult to collect crop residues. I also show that farmers' income increases in response to the plant openings.

I conduct additional tests to rule out some alternative explanations. First, the findings are not caused by structural transformation induced by a plant. Following the arrival of a plant, I find no change in agricultural land or other dimensions of agricultural production such as crop types and agricultural productivity. I also find no change in the industrial sector, including the very small firms, as measured by the number and size of industrial firms. Second, the findings are not caused by migration due to fear of possible pollution from a plant. I focus on areas downwind of a plant that are perceived to be affected more by the plant and do not find a larger effect. I also show that the effects in areas with access to transport networks, where it is easier to move out, are similar to other areas. Third, the findings are not caused by enhanced inspection by local officials around a plant, as the arrival of a plant in an area could signal local officials' attention paid to that area. I do not find a larger effect in areas with greater penetration of local officials. I also focus on areas where local officials have stronger promotion incentives, as pollution control is considered in promotion. I do not find a larger effect.

For the analysis of no burning zones, I justify the validity of the RD strategy in the following ways. First, I check the relevant documents and show that the drawing of no burning zones is a uniform and top-down decision made by the central government, which is uncorrelated with local economic and environmental conditions. Second, I check the preexisting covariates and find no discernible jumps at the zone boundaries. Third, I test possible selective sorting of farmers, firms and airports. The results either suggest no selective sorting or are against the main findings (i.e., some types of sorting would lead to a significant reduction in fires, whereas I find a small and insignificant reduction). Therefore, the main findings are unlikely to be driven by various forms of sorting. Fourth, I conduct a difference-in-discontinuities analysis exploiting within grid cell variation to rule out confounding differences in areas across the zone boundaries. Finally, I complement the RD design with a DID strategy that considers areas far away from the boundaries to strengthen external validity.

Using data on 205 airports, I find little evidence that no burning zones could reduce agricultural fires. In the preferred specification that uses a linear RD polynomial in distance to the boundaries and controls for airport × boundary segment fixed effects, the estimated coefficients only account for 1% of the sample mean, and are statistically insignificant. The results are robust to alternative bandwidths, fixed effects, samples, polynomial orders, and inclusion of covariates. I also use a two-dimensional RD specification using the longitude and latitude of each grid cell as polynomials, and the results barely change. I next check whether such insignificant correlation between no burning zones and agricultural fires masks some heterogeneity, by conducting RD estimation separately for each year and each airport. The effects are still small and statistically insignificant for most years and airports.

I further test whether the effects are significant in areas that are supposed to have more stringent regulatory enforcement. I look at three types of areas. The first type is areas upwind of the airport. Due to safety concerns, local officials should pay greater attention to such areas. The second type is areas in provincial cities. These cities are more developed and have higher political status. Hence, local officials should have stronger incentive to control pollution for both economic and political concerns. Third, areas where the city party secretary has stronger promotion incentive. As pollution control is considered as part of the criteria for promotion, regulatory enforcement should also be stronger in such areas. I still cannot find significant effects in all these three types of areas. In addition, I also check whether the zones shift the burning from daytime to nighttime, which may still be meaningful due to more flights in the daytime. I do not find any supporting evidence.

This paper is primarily related to a nascent literature showing the causal effects of environmental policies on agricultural fires, a typical type of pollution in the rural areas (He et al., 2020a). As shown by prior work, agricultural fires could deteriorate air quality (Guo, 2020), affect infant and adult health (Rangel and Vogl, 2019; He et al., 2020a), and harm cognitive performance (Graff Zivin et al., 2020). Moreover, agricultural fires could generate large spillover effects in the sense that the pollutants could travel a long distance of about several hundreds of kilometers (Guo, 2020). Apart from such direct effects on human health, smokes from intensive burning could also threaten traffic safety, creating indirect effects on human health (Sager, 2019). Given such substantial health costs, finding the most effective way to reduce agricultural fires has significant welfare implications. In a closely related

study, He et al. (2020a) show that straw-recycling subsidy at the provincial level can reduce agricultural fires. My paper differs in two aspects. First, the straw-recycling subsidy studied in He et al. (2020a) is similar to a traditional conditional cash transfer, which may entail a nontrivial enforcement cost; the plants studied in my paper, however, work more like a market device to automatically incentivize farmers to reduce burning. Second, in addition to the incentive-based policy, I study another commonly used command and control policy, which is unexamined in He et al. (2020a). Using micro data and hyperlocal variation in policy exposure, I am able to test the causal effects of these policies and show that in contrast to command and control, resorting to economic incentives could significantly reduce agricultural fires.

This paper is also broadly related to a growing literature studying the effectiveness of environmental policies in developing countries. A common finding in this strand of literature is that command and control policies in these countries tend to fail due to weak regulatory enforcement (Davis, 2008; Duflo et al., 2013, 2018; Oliva, 2015; Souza-Rodrigues, 2019). However, a few studies do find a significant effect of command and control policies. For example, Greenstone and Hanna (2014) demonstrate the effectiveness of the regulation in India targeting air quality. Harrison et al. (2015) focus on firm-level outcomes in India and find both the incentive-based policy and the command and control policy could work for pollution abatement. Some studies based on China also find significant reductions in pollution from command and control policies (Tanaka, 2015; Viard and Fu, 2015; He et al., 2020b). Apart from the mixed findings, a common feature of these studies is that they focus on pollution in the urban areas, which differ from rural areas in many aspects such as social norms, enforcement costs, and marginal willingness to pay for environmental quality. Therefore, whether the existing results apply to rural areas is unclear. In terms of the rural setting, my paper is closely related to Souza-Rodrigues (2019), who shows through a structural estimation that economic incentives, rather than command and control, can reduce deforestation in the Amazon. However, as deforestation mainly happens in areas with a shortage of agricultural land and weak state capacity, it is still unclear whether such results could apply to my setting.

The rest of this paper proceeds as follows. Section 2 discusses the institutional background. Section 3 describes the data and presents descriptive statistics. Sections 4 and 5 examine the effects of biomass power plants and no burning zones on agricultural fires, respectively. Section 6 discusses some additional issues and Section 7 concludes.

2. Institutional background

2.1. Agricultural fires across the world and in China

Across various farming systems, burning is one of the most popular ways to remove the cumbersome crop residues after harvest. Farmers view burning as not only a convenient way to clean the land for subsequent cultivation but also a way to nourish soils and kill pests, although burning is demonstrated to be harmful to soil quality (Villar et al., 2004). The pollutants from intense burning in a short period could release various pollutants (Chen et al., 2017a). The most typical pollutant is particulate matter (e.g., $PM_{2.5}$), which could stay in the air for a long time and cause respiratory and cardiovascular syndromes, among others (Dominici et al., 2006). Besides, the fine particles could also cause haze and reduce visibility (Cheng et al., 2014). Other pollutants include CO, VOCs, NOx, SO_2, and other compound toxic substances. In sum, burning crop residues not only affects human health but also generates environmental damages.

Despite the rapid economic growth in recent years, China remains a largely rural country and is currently the largest burner of crop residues in aggregate terms. There are several features specific to China that could explain the burning. First, there is a very short window for removing crop residues between one harvest and the next sowing, typically one or two weeks. Given such short period and lack of labor force due to migration of young people,[4] burning seems to be the most efficient and economic way to prepare for sowing. Second, crop residues were traditionally used as fuel for cooking and were collected and used gradually. The recent spread of natural gas among the rural areas makes the crop residues quite useless, leading to intense field burning in a short time. Third, in China each farmer only owns a small piece of land. The wide dispersion of farming makes it hard for local officials to monitor burning.

2.2. Biomass power plants

Biomass power plants use agricultural waste or other organic waste to generate energy, either directly through burning or indirectly through gasification. After the 1973 oil crisis, Denmark first proposed this technology as a safeguard against potential oil supply crisis, as it relied heavily on oil. China was quite late in adopting this technology and it was not until the early 2000s that biomass power plants started to appear. In the Renewable Energy Law enacted in 2005,[5] biomass energy was listed as a promising energy source. The National Development and Reform Commission then launched various policies to subsidize biomass power plants to boost their development. As said in the law, the main purpose of promoting this type of energy is to expand energy sources and decrease the reliance on fossil energy. Since then, biomass power plants have grown quite rapidly.

Compared to open burning of crop residues, biomass power plants are much cleaner in the sense that they could use dedusting, desulfurization and denitrification equipment to absorb most of the pollutants. Apart from burning, these plants could also transform the residues into gas, which generates fewer pollutants after burning. In this paper, I focus on the plants that mainly use agricultural waste as inputs.[6] For these plants, they typically use crop residues from surrounding farms.[7] Although farmers are paid for the crop residues, they need to collect and package the residues from the farm, which is a laborious process. The price of the crop residues is quite low, making some farmers view it unworthy to collect and sell the crop residues to plants far away. That being said, the revenue from crop residues is still nontrivial and could be attractive to some farmers, as shown in Appendix F.1.

2.3. No burning zones

In China, airports are typically located in the suburbs and surrounded mostly by farms. The smoke from agricultural fires, which is much more intense and dangerous than naturally formed fog, could severely affect the take-off and landing of airplanes.[8] On May 13th, 1998, due to intense agricultural fires, Shuangliu Airport in southwest China was forced to close for three times in one day, leading to the delay, return, or diversion of dozens of planes.[9] In an effort to protect air quality and secure airplane safety, the Ministry of Environmental Protection proposed a ban in 1999 on burning crop residues within 15 km of airports.[10] To make the ban public, local officials were required

[4] Migration here refers to short-term change of residence, as permanent or long-term change is hard for farmers due to the *hukou* system.

[5] See http://www.gov.cn/ziliao/flfg/2005-06/21/content_8275.htm for the relevant Chinese document.

[6] Another type of plant uses mainly organic waste from the urban areas as inputs, and is typically located in the urban areas.

[7] A typical biomass power plant usually has a storage center that stores residues collected during the harvest season to smooth the supply of residues over the year. See http://www.nantong.gov.cn/jcms/jcms_files/jcms1/web14/site/attach/0/160429135950571.pdf.

[8] See http://news.sina.com.cn/c/2002-10-09/0853759792.html for a survey with officers from an airport in China who expressed concerns about the threats from agricultural fires.

[9] See https://cn.govopendata.com/renminribao/1998/5/19/4/#1120306.

[10] See http://www.mee.gov.cn/gkml/zj/wj/200910/t20091022_171920.htm.

to print out copies of the ban and disseminate them to farmers. Moreover, local officials should patrol these zones routinely to discipline or prosecute the violators.

Several features make such zones unique and suitable to address identification challenges. First, they are formulated uniformly from the top and therefore are unlikely to be correlated with local economic and environmental conditions. Second, they are drawn artificially along the 15 km-radius circle surrounding each airport and do not overlap with other important boundaries, avoiding the compound treatments issue faced by many geographic regression discontinuity designs utilizing administrative boundaries (Keele and Titiunik, 2015). Third, due to the household registration system (*hukou*), it is quite hard for farmers to migrate, largely alleviating the selective sorting issue in many regression discontinuity designs (McCrary, 2008).

In addition to the no burning zones around airports, the 1999 regulation also banned fires in areas within 1 or 2 km of major transportation networks. Local governments were also entitled to set up additional zones. It would also be interesting to examine the impact of these additional zones. However, the zones along major transportation networks may be too narrow to fit a smooth function in RD design. Furthermore, areas closer to major transportation networks could differ from areas farther away ex ante, and such difference may be discontinuous at the zone boundaries given that the zones are too narrow. Finally, even if all factors are ex ante continuous, it is hard to separate the effect of the zones from other ex post changes created by proximity to transportation networks. For the local government-specified zones, they are also not utilized in my analysis for several reasons. First, there is little exact delineation of these zones. In very few cases where there is a well-specified zone, the border of such zone usually coincides with major roads or administrative boundaries, making it difficult to separate the role of no burning zones from the role of roads or other boundaries. Second, as these zones are mostly in the urban areas, there may not be enough agricultural land around the zone boundaries. Therefore, one may find null effects simply due to low statistical power. Third, as these zones are determined at the discretion of local governments, they have strong political incentives to place the zones in areas with lower burnings. In sum, it is difficult to draw credible causal inference from these additional zones.

3. Data and summary statistics

The data in this paper are aggregated at the 1 km × 1 km grid cell level. To this end, I create a fixed grid with cells of 1 km × 1 km and then match all the data to this grid. I use this level of aggregation for several reasons. First, both biomass power plants and no burning zones could affect areas spanning multiple administrative units, making it hard to determine the treatment unit and control unit. For example, if a plant is located near the border of a city, then it is unreasonable to assign only this city as the treatment city. Second, as the boundary of each no burning zone is a 15 km-radius circle around the airport, then using such grid cells allows me to generate enough observations around the treatment threshold to conduct RD estimation. Third, using such disaggregated analysis allows me to better measure the geographic characteristics and control for confounding factors using grid cell fixed effects. More importantly, I could also control for the selective arrival of a plant in an area by including plant × year and city × year fixed effects.

3.1. Main data

Agricultural fires data Data on fires are generated by the NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) aboard the Terra and Aqua satellites, available from 2001 onwards. I use the data from 2001 to 2016. The two satellites pass over the equator for four times each day (10:30 am, 10:30 pm, 1:30 pm, and 1:30 am) and observe almost every area on the earth for at least 4 times due to overlapping orbits. The MODIS sensor detects fires using a contextual algorithm utilizing the strong emission of mid-infrared radiation from fires and reports the longitude and latitude of a fire. Giglio et al. (2003). Depending on the observing conditions, the minimum detectable fire size ranges from 50 to 1000 m^2. As the average size of cropland in China is much larger (typically about several thousand m^2), the sensor should be able to detect most agricultural fires. As the focus of this paper is agricultural fires, I next match the fire data with land cover data from the Copernicus Land Monitoring Service (CLMS), to determine fires occurring in cropland. The CLMS classifies global land surface into several classes, including cropland and settlement land. The data are available annually from 1992 onwards. I define a fire as agricultural fire if the location of the fire is inside cropland.

As is common in most satellite observations data, the fire observations data are not immune to potential measurement errors. The first type of measurement error comes from varying observing conditions. That is, the detection could be affected by factors like cloud cover, land surface temperature, and amount of smoke. This type of error is plausibly random and therefore can be viewed as classical measurement error. The results are still unbiased, but the precision will be lower. The second type of measurement error comes from the limited number of observations each day, leading to an under-detection of fires. For example, farmers in certain area may choose to burn when the satellites are not passing over that area, either due to behavioral changes or fear of being detected by relevant authorities. However, as I include the plant × year, city × year, or airport × year fixed effects, depending on the specifications, such under-detection should be largely controlled by these fixed effects.

Biomass power plants data Data on biomass power plants are obtained from a Chinese consulting firm. The firm collected the information on all the active plants in 2016 and verified the information through multiple ways including phone calls. The information includes plant name, detailed address, year of opening, and main business. I use the APIs from Baidu Maps, the Chinese equivalent of Google Maps, to geocode the plants to get the longitudes and latitudes.[11] There are mainly two types of plants: one uses crop residues as inputs and the other uses organic waste from the urban areas as inputs. I keep only those plants that use crop residues as inputs. There are 253 such plants. I validate such number using an official report and discuss the potential bias in Appendix E.1. These plants are basically located in the rural areas and on average over 70% of the surrounding land is cropland. To match the fire data that are available only from 2001, I drop plants established before 2002 to have at least one year of fire data before the opening of any plant. Similarly, I also drop plants established after 2015 to have at least one year of fire data after the opening of any plant.[12] Then I am left with 249 plants. Using each of the remaining plants as a center, I draw a circle with a 15 km radius. I drop those plants with overlapping circles, as in this case it would be difficult to define the treatment and control areas. This step excludes about 24% plants and I am left with 190 plants. Appendix E.3 shows that such sample selection has little impact on my findings. The spatial and temporal distribution of these plants are shown in Fig. 2. The plants are concentrated in the major grain-producing provinces such as Shandong, Henan, and Heilongjiang, and increase sharply after the introduction of the Renewable Energy Law in 2005.

[11] The longitudes and latitudes obtained directly from the *geocoding* API from Baidu Maps use the GCJ02 coordinate system, which is censored as it adds some offsets about several hundred meters to the widely used WGS84 coordinates. To match with other geographic data that use the WGS84 coordinates, I rely on the *coordination conversion* API to iteratively convert the GCJ02 coordinates into WGS84 coordinates. The procedure is described in detail in Appendix E.2.

[12] This step drops very few plants and results are similar if I do not drop these plants.

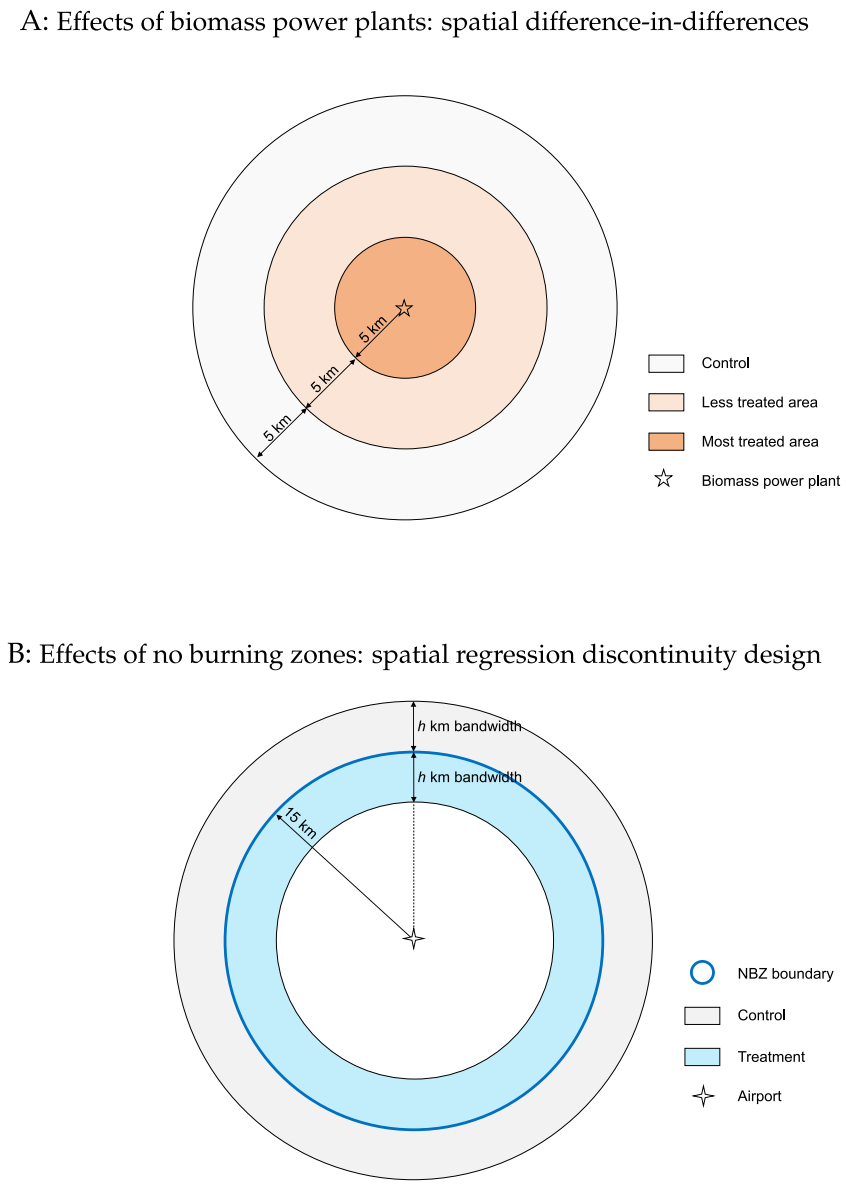


Fig. 1. Illustration of identification strategy.
Notes: The unit of observation is 1 km × 1 km grid cells. This figure illustrates the construction of treatment and control groups used in testing the effects of biomass power plants and no burning zones on agricultural fires, respectively.

No burning zones data To construct the no burning zones, I first get the list of civic airports from the Great Circle Mapper.[13] For confidentiality reasons, data on military airports are not available. I use the 205 active civic airports at the end of 2016. For each airport, I have the longitude and latitude and the year of opening. I then draw a circle of 30 km radius around each airport. Because these 30 km-radius circles do not overlap with each other, all the airports are kept. All the areas inside the circle constitute the sample for RD design. The treatment threshold is a circle of 15 km radius.[14] Inside this circle, agricultural fires are banned, and outside, not. The spatial and temporal distribution of these airports are shown in Fig. 2. As can be seen, some airports are located in provinces where agriculture is not well developed, such as Qinghai and Xizhang province in southwest China. Therefore, areas around these airports may not be suitable for studying agricultural fires. I address such concern by eliminating airports around which very little cropland exists in robustness checks.

3.2. Other data

Below I list some data that are used either for mechanism tests or heterogeneity analysis in the main text. For some other data used in the appendix, I will describe them where they first appear.

Annual survey of industrial firms (ASIF) data I use the ASIF data from 1998 to 2007, collected by the National Bureau of Statistics of

[13] See www.gcmap.com for details.

[14] When drawing the 15 km-radius circles, an airport is treated as a point using its location shown on the map, instead of a polygon. This is consistent with the procedure used by the Ministry of Environmental Protection of China to determine whether a fire is within the no burning zones (Fang et al., 2006). To alleviate concerns that airports may expand and slightly change their locations on the map, I show in Appendix Table A23 that the results are robust when using a "donut hole" RD specification to exclude areas close to the 15 km-radius circles.

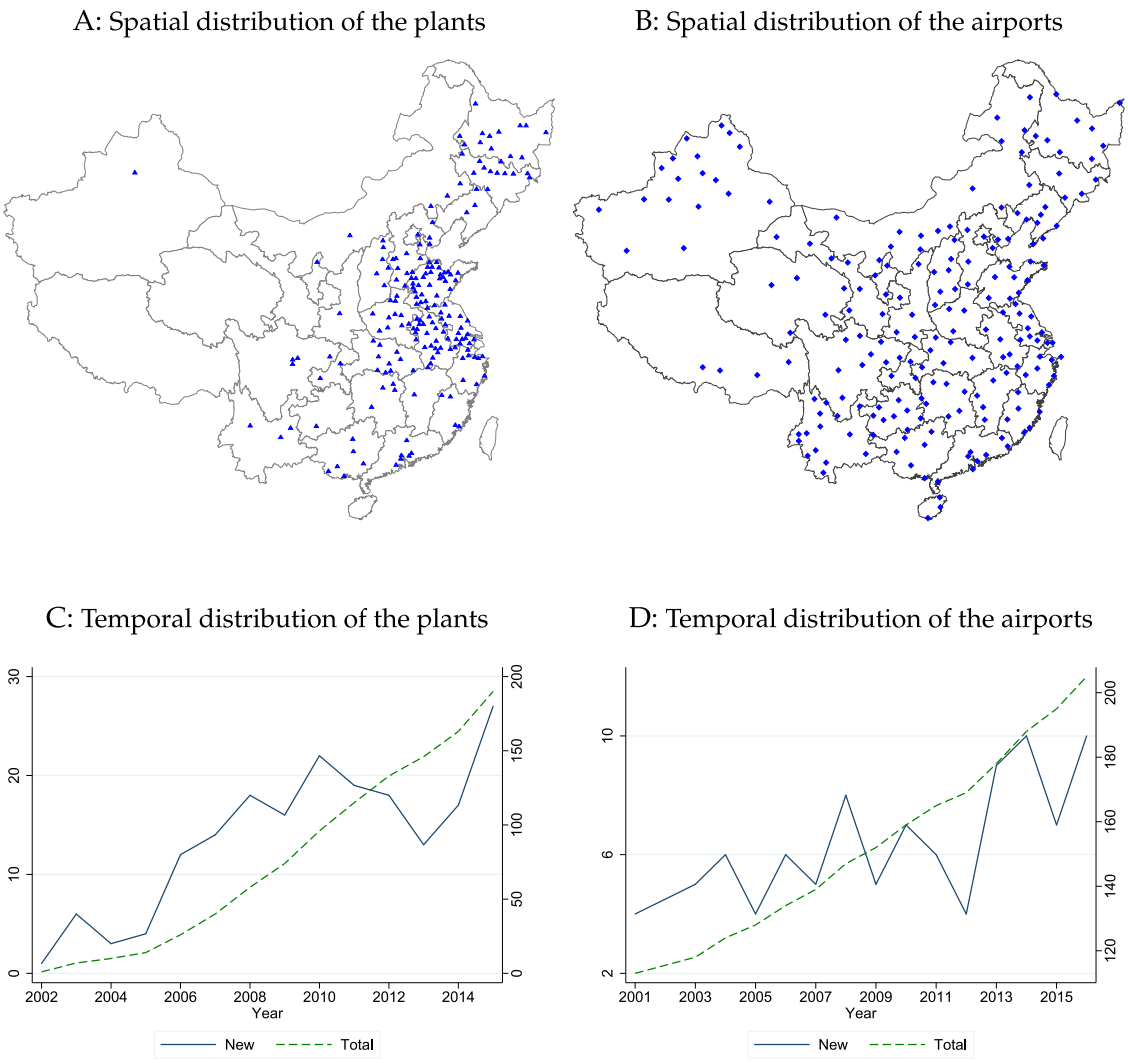


Fig. 2. Distribution of biomass power plants and airports.

China, to construct measures reflecting the development of the industrial sector in each grid cell. The data include private industrial firms with annual sales over 5 million *yuan* (about 0.78 million USD) and all state-owned industrial enterprises (SOEs). The data also have detailed address information for each firm, which enables me to geocode the location. As the data contain some outliers, I follow standard procedures used in the literature to clean the data (Brandt et al., 2012).

Geography and transportation data Data on land cover from the CLMS are used to construct the share of cropland and settlement land in each grid cell. Data on topography from the NASA's Shuttle Radar Topography Mission (SRTM) are used to construct the slope, elevation, and ruggedness of each grid cell, following the method used by Nunn and Puga (2012). Data on river networks from the National Geomatics Center of China are used to construct the number of rivers in each grid cell and the distance from the centroid of each grid cell to the nearest rivers. Data on transportation networks come from the Geographic Data Platform maintained by Peking University, including railways, national roads, and expressways. The data are available only for 2000.

Wind direction data I obtain wind direction data from the National Oceanic and Atmospheric Administration (NOAA) to gauge how frequent a grid cell is downwind of a plant (or upwind of an airport). The wind direction is observed at three-hour frequencies by ground-based weather stations and then averaged to the daily level using the vector decomposition method.[15] I assign the daily wind direction to a plant (or an airport) using the observations from the nearest weather station and drop plants (or airports) with no stations within 100 km. I then calculate the percentage of days in a year that a grid cell spends downwind of a plant (or upwind of an airport), where downwind (or upwind) is defined following the literature (Rangel and Vogl, 2019; Graff Zivin et al., 2020; He et al., 2020a) and also illustrated in Appendix Figure A1.

City leader data Data on the top 2 leaders in a city (the party secretary and mayor) collected from multiple sources are used to construct measures of political incentives. The sources include the provincial and city statistical yearbooks, Baidu Baike (China's Wikipedia) and Wikipedia. The data contain detailed information about the resume of each leader from 2001 to 2016, and are cross-validated whenever possible.

3.3. Descriptive statistics

Appendix Table A1 contains descriptive statistics on the final sample used to analyze the effects of biomass power plants. That is, areas within 15 km of a plant. Appendix Table A2 contains descriptive statistics on the final sample used to analyze the effects of no burning zones based on a 4 km bandwidth, which is roughly the average

[15] See http://www.webmet.com/met_monitoring/622.html.

bandwidth across different variables. Most of the time-varying variables are measured from 2001 to 2016. For grid cells around a plant, the average number of agricultural fires is 0.022, and the probability of observing at least one agricultural fire is 0.017. For grid cells around the no burning zone boundaries, the average number of agricultural fires is 0.011, and the probability of observing at least one agricultural fire is 0.009.

Fig. 3 gives a description of the evolution of agricultural fires around biomass power plants and no burning zone boundaries. At the plant level, the number of agricultural fires is steadily increasing (Panel A). Panel C gives the average number of agricultural fires at the grid cell level, conditional on treatment status. Overall, there is an increasing trend across all three groups, although such trend seems to be reversed after 2013.[16] Between groups, there is no clear difference. Panel B and D show the evolution of agricultural fires around the no burning zone boundaries. The pattern is similar. Appendix Figure A2 gives a comparison of agricultural fires before and after a plant opening for different treatment groups. In Panel A, the most treated areas seem to have the largest number of agricultural fires before the opening (although the difference across groups is small), and such pattern is reversed after the opening. A somewhat similar pattern is also found in Panel B where I examine the probability of observing at least one agricultural fire. Such pattern sheds some light on the effects of biomass power plants on reducing burning, although it is just correlation.

4. The effects of biomass power plants

4.1. Model specification

I evaluate the effects of biomass power plants on agricultural fires using a geographic difference-in-differences (DID) strategy (Currie and Walker, 2011; Currie et al., 2015). As illustrated in Fig. 1, the treatment and control groups are defined based on the proximity of each area to the nearest plant. The specification is as follows:

$$Y_{ijt} = \delta_i + \lambda_{jt} + \beta_1 Distance_{ij}^{<5\text{ km}} \times Post_{jt} + \beta_2 Distance_{ij}^{5-10\text{ km}} \times Post_{jt} + \epsilon_{ijt} \quad (1)$$

where the unit of observation is 1 km × 1 km grid cells and the sample includes grid cells within 15 km of a plant. i, j, and t denotes grid cell, plant, and year, respectively. Y_{ijt} denotes measures of agricultural fires in grid cell i around plant j at year t, either in terms of incidence (presence of at least one fire) or in terms of intensity (number of fires). δ_i denotes grid cell fixed effects, controlling for time-invariant regional characteristics at the grid cell level such as topography. λ_{jt} denotes plant×year fixed effects, controlling for all factors, including those time-varying, affecting all the grid cells around a plant such as weather conditions, plant performance, and local economic prosperity. $Distance_{ij}^{<5\text{ km}}$ is an indicator equal to 1 if the distance from the centroid of grid cell i to plant j is smaller than 5 km and 0 otherwise; $Distance_{ij}^{5-10\text{ km}}$ is an indicator equal to 1 if the distance from the centroid of grid cell i to plant j is between 5 km and 10 km and 0 otherwise. $Post_{jt}$ is an indicator equal to 1 for years after the opening of plant j and 0 otherwise. As I do not observe the exact month of opening, in Appendix E.4 I show that the results are robust when excluding years around a plant opening or assigning an expected treatment value between 0 and 1 to $Post_{jt}$ in the year of a plant opening following Currie et al. (2015).

The two indicators $Distance_{ij}^{<5\text{ km}}$ and $Distance_{ij}^{5-10\text{ km}}$ denote the most treated areas and the less treated areas, respectively. Choosing such areas as treatment groups is motivated by the fact that crop residues bear high transportation costs and low prices. Anecdotal evidence shows that farmers typically are only willing to send them to plants several kilometers away. Therefore, I choose the areas within 10 km of a plant as the treatment groups and the areas between 10 km and 15 km as the control group. I further bolster such choice by estimating a flexible variant of the baseline equation with many 5-km distance bins in Appendix A. To circumvent choosing treatment thresholds, I also use continuous treatment intensity with the two binary indicators replaced by continuous distance to a plant in robustness checks, and use least cost distance calculated based on topography in Appendix C.[17] The coefficients of interest are β_1 and β_2, which are standard DID estimators. Specifically, β_1 measures the change in agricultural fires in areas within 5 km of a plant after its opening, relative to areas 10 km away; β_2 measures the change in agricultural fires in areas within 5–10 km of a plant after its opening, relative to areas 10 km away.

In the baseline specification, I cluster the standard errors at the plant level to account for temporal correlation that is severe in DID estimation (Bertrand et al., 2004; Cameron and Miller, 2015), as well as spatial correlation across grid cells around the same plant. Given the nature of the data, especially the high spatial resolution, controlling for spatial correlation is important. I establish robustness of the results to adjusting the standard errors in two more conservative ways: (1) clustering at the 2° longitude × 2° latitude level[18]; (2) allowing for arbitrary correlation across spatially adjacent observations (Conley, 1999; Hsiang, 2010).

Recent econometric literature shows that the estimates from staggered DID designs are not easily interpretable (e.g., Callaway and Sant'Anna, 2020; de Chaisemartin and d'Haultfoeuille, 2020; Goodman-Bacon, 2021). In Appendix B, I show that this is not a major concern in my setting; furthermore, the results are similar when I account for such issue using a more interpretable estimator following Callaway and Sant'Anna (2020).

4.2. Identification concerns

The causal interpretation of the effects found in Eq. (1) relies on the common trends assumption. That is, agricultural fires in the treatment group and the control group should grow in parallel in the absence of the plant. The major threat to this assumption is that the arrival of a plant in certain area is not random and may be correlated with certain local conditions. The grid cell fixed effects and plant × year fixed effects already take into consideration time-invariant factors at the grid cell level and time-varying factors around the plant that could drive the arrival of the plant. For most plants, both the treatment areas and control areas are also nested in the city where the plant is located. Therefore, the plant × year fixed effects could largely control for city-level factors that could lead to selection issue. I also control for both plant × year fixed effects and city × year fixed effects in robustness checks.

The remaining concerns are some time-varying factors at the grid cell level that may affect the arrival of a plant and also fires. One may argue that a plant could be placed in areas with growing agricultural fires. Conceptually, as the plants are mainly used to expand energy source instead of reducing agricultural fires, this issue is less severe. Nevertheless, I adopt a number of strategies to address such concerns.

First, I use a more demanding specification to allow for both differential linear trends in each grid cell and arbitrary trends in each grid

[16] This reversal is consistent with China's significant efforts on combating air pollution since 2013 (Greenstone et al., 2021).

[17] Additionally, one may also want to use travel distance from a grid cell to the plant. However, data on roads in the rural area are essentially unavailable. Moreover, such travel distance would be endogenous and may also capture the confounding effect of transportation connection.

[18] For anywhere on the earth, 1° latitude ≈ 111 km. For China, 1° longitude ≈ 70–100 km (larger when one moves southward).

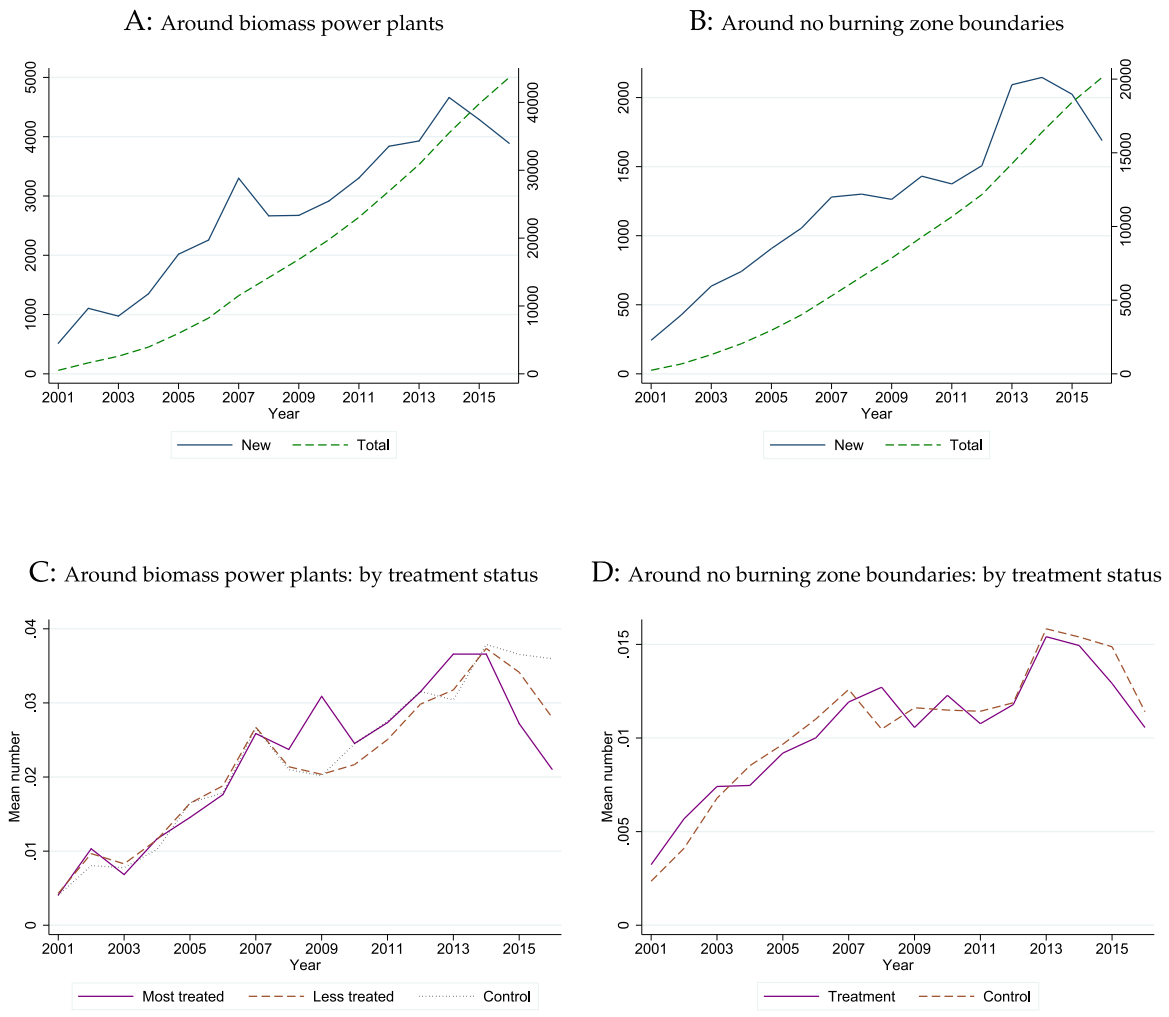


Fig. 3. Agricultural Fires around biomass power plants and no burning zone boundaries.
Notes: Panel A shows the number of agricultural fires within 15 km of biomass power plants. Panel B shows the number of agricultural fires within 4 km of the no burning zone boundaries. Panel C and D break down them by treatment status.

cell depending on some geographic characteristics. The specification is as follows:

$$Y_{ijt} = \delta_i + \lambda_{jt} + \delta_i \times Year_t + \mathbf{X}_i \times \theta_t + \beta_1 Distance_{ij}^{<5\text{ km}} \times Post_{jt} + \beta_2 Distance_{ij}^{5-10\text{ km}} \times Post_{jt} + \varepsilon_{ijt} \quad (2)$$

where $Year_t$ denotes the number of years elapsed since the initial year. The term $\delta_i \times Year_t$ thus allows for a differential linear time trend in each grid cell. Appendix Table A4 shows similar results for higher-order time trends. $\mathbf{X}_i$ is a set of geographic characteristics including longitude, latitude, slope, ruggedness, elevation, distance to rivers, and number of rivers. By interacting these characteristics with year fixed effects θ_t, I allow the grid cells to evolve differentially depending on these characteristics. Conditional on the previous fixed effects and such flexible trends, the common-trends assumption is more likely to hold.

Second, I adopt an event study specification to directly test whether there exist any diverging pre-trends:

$$Y_{ijt} = \delta_i + \lambda_{jt} + \delta_i \times Year_t + \mathbf{X}_i \times \theta_t + \sum_k \beta_1^k Distance_{ij}^{<5\text{ km}} \times Post_{jt}^k + \sum_k \beta_2^k Distance_{ij}^{5-10\text{ km}} \times Post_{jt}^k + \varepsilon_{ijt} \quad (3)$$
$$s.t. \quad k \in \{-5, -4, -3, -2, 0, 1, 2, 3, 4, 5+\}$$

where I replace the $Post_{jt}$ indicator in the previous equation with a set of indicators denoting different years relative to the year of a plant opening. Specifically, $Post_{jt}^k$ equals to 1 if year t is the kth year relative to the opening of plant j. Negative ks denote years prior to the opening. $k = 5+$ denotes the fifth year after a plant opening and all subsequent years. $k = -5$ denotes the fifth year before a plant opening and all prior years. The year prior to a plant opening is omitted as the reference year. Hence, β_1^k measures the change in agricultural fires in areas within 5 km of a plant between the kth year relative to its opening and the year prior to its opening, compared to areas 10 km away; β_2^k measures the change in agricultural fires in areas within 5–10 km of a plant between the kth year relative to its opening and the year prior to its opening, compared to areas 10 km away. If the β^ks are small and statistically insignificant for $k \leq 0$, then the common trends assumption is likely to hold.

Third, I conduct two placebo tests as follows

$$Y_{ijt}^{placebo} = \delta_i + \lambda_{jt} + \delta_i \times Year_t + \mathbf{X}_i \times \theta_t + \beta_1 Distance_{ij}^{<5\text{ km}} \times Post_{jt} + \beta_2 Distance_{ij}^{5-10\text{ km}} \times Post_{jt} + \varepsilon_{ijt} \quad (4)$$

$$Y_{ijt} = \delta_i + \lambda_{jt} + \delta_i \times Year_t + \mathbf{X}_i \times \theta_t + \beta_1 Distance_{ij}^{<5\text{ km}} \times Post_{jt}^{placebo} + \beta_2 Distance_{ij}^{5-10\text{ km}} \times Post_{jt}^{placebo} + \varepsilon_{ijt} \quad (5)$$

First, in Eq. (4), I replace the dependent variables in previous equations with measures of non-agricultural fires $Y_{ijt}^{placebo}$, which are unlikely to be affected by biomass power plants. If the common trends assumption holds, then by construction one should not find a significant

Table 1
The effect of biomass power plants on agricultural fires.

Dep. var. =	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Number of fires				Presence of fires			
<5 km of the plant × Post	−0.0054**	−0.0085***	−0.0088***	−0.0127***	−0.0033**	−0.0052***	−0.0055***	−0.0065***
	(0.0025)	(0.0020)	(0.0020)	(0.0028)	(0.0014)	(0.0012)	(0.0012)	(0.0015)
5–10 km of the plant × Post	−0.0027*	−0.0049***	−0.0049***	−0.0068***	−0.0018**	−0.0018**	−0.0019**	−0.0027***
	(0.0015)	(0.0015)	(0.0015)	(0.0018)	(0.0008)	(0.0008)	(0.0008)	(0.0009)
<5 km of the plant	0.0014				0.0001			
	(0.0022)				(0.0011)			
5–10 km of the plant	0.0003				0.0012**			
	(0.0011)				(0.0006)			
Grid cell FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Plant × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geo. controls × Year FE	No	No	Yes	Yes	No	No	Yes	Yes
Grid cell trends	No	No	No	Yes	No	No	No	Yes
Cluster level	Plant	Plant	Plant	Plant	Plant	Plant	Plant	Plant
Observations	1,986,576	1,986,576	1,986,576	1,986,576	1,986,576	1,986,576	1,986,576	1,986,576
R-squared	0.0187	0.2729	0.2731	0.5196	0.0428	0.1615	0.1617	0.2378
Mean of dep. var.	0.0220	0.0220	0.0220	0.0220	0.0174	0.0174	0.0174	0.0174

Notes: The unit of observation is 1 km × 1 km grid cells. This table shows that biomass power plants can reduce nearby agricultural fires, using a spatial difference-in-differences strategy. *Presence of fires* is an indicator for observing at least one fire. *<5* km *of the plant* and *5–10* km *of the plant* are indicators for grid cells within 5 km and 5–10 km of the plant, respectively. *Post* is an indicator for years after the plant opening. Geographic controls include longitude, latitude, slope, ruggedness, elevation, distance to rivers, and number of rivers. Standard errors in parentheses are clustered at the plant level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

effect. Second, I randomly permutate the timing of the plant openings to generate a large number of placebo treatments reflected by the indicator $Post_{jt}^{placebo}$ in Eq. (5). Then if the results are simply driven by differential trends, then one should see similar effects among many placebo estimations.

I address a number of other selection issues in Appendix D. Specifically, I show that the results hold if I directly control for possible selection into areas with high amount of crop residues or concentrated land ownership that may correlate with fires; I also show that the results are not sensitive to correcting for selection on unobservables using the method proposed by Oster (2019), which assumes that selection on observables is informative about selection on unobservables.

4.3. Baseline results

Average effect Table 1 shows the DID estimation of the effects of biomass power plants on agricultural fires. Columns (1)–(4) examine the number of fires. Column (1) only includes plant × year fixed effects. The coefficients on the two treatment indicators *< 5* km *of the plant* and *5–10* km *of the plant* are small and statistically insignificant, meaning that the average number of fires are similar across the most treated areas, the less treated areas and the control areas before the opening of a plant. The coefficients on *< 5 km of the plant × Post* and *5–10* km *of the plant × Post* are −0.0054 and −0.0027, respectively, and both are statistically significant. Compared to the sample mean (0.022), these estimates suggest that the opening of a plant accounts for a 24.5% drop and a 12.3% drop in the number of fires in the most treated areas and the less treated areas, respectively.[19]

Column (2) adds grid cell fixed effects to control for time-invariant characteristics at a highly disaggregated level. The results show a larger and statistically significant reduction in the number of fires in both the most treated areas and the less treated areas. Column (3) adds a set of geographic controls interacted with time fixed effects to allow for different areas to evolve differently according to their geographic characteristics. The results barely change. Column (4) further allows different areas to have different trends by including grid cell-specific linear trends. This specification together with that in column (3) could alleviate the concerns that the results are capturing the differential trends across areas. Compared to the sample mean, this most restrictive specification in column (4) implies a 57.7% drop and a 30.9% drop in the number of fires in the most treated areas and in the less treated areas, respectively. Columns (5)–(8) replace the number of fires with a binary indicator denoting observing at least one fire and get similar patterns to those in columns (1)–(4). Take the most restrictive specification in column (8) as an example, compared to the sample mean, the opening of a plant leads to a 37.4% drop and a 15.5% drop in the probability that fires appear in the most treated areas and the less treated areas, respectively.

Event study To check whether there exist differential pre-trends, I use the event study specification, Eq. (3), in which I interact the treatment indicators with a set of indicators denoting years before or after a plant opening. The year prior to the opening is omitted as the reference year. The results are visualized in Fig. 4. Panel A and B only include grid cell fixed effects and plant × year fixed effects. The difference in agricultural fires between the treatment areas and the control areas are small and statistically insignificant in years before the opening, and are larger and statistically significant in years after the opening. Panel C and D further add geographic controls interacted with year fixed effects and grid cell-specific linear trends to control for possible differential trends across areas. The patterns are similar to those shown in Panel A and B. Overall, these patterns confirm that the most treated areas, the less treated areas, and the control areas share similar trends in agricultural fires before the plant opening, and that these areas start to diverge only after the plant opening.

4.4. Robustness

Continuous treatment intensity Results are similar when I replace the binary treatment indicators with continuous treatment intensity, which is the distance from each area to the plant. This specification could alleviate concerns that a plant could also affect areas 10 km away. I estimate the following equation:

$$Y_{ijt} = \delta_i + \lambda_{jt} + \delta_i \times Year_t + \mathbf{X}_i \times \theta_t + \beta Distance_{ij} \times Post_{jt} + \varepsilon_{ijt} \quad (6)$$

[19] There is a concern that the smaller coefficient for the less treated areas may be driven by the initial difference in fire activity across different areas. To alleviate this concern, I use the logarithm of one plus the number of fires as the outcome variable. Appendix Table A5 shows that the coefficient (and economic magnitude) for the less treated areas is still much smaller.

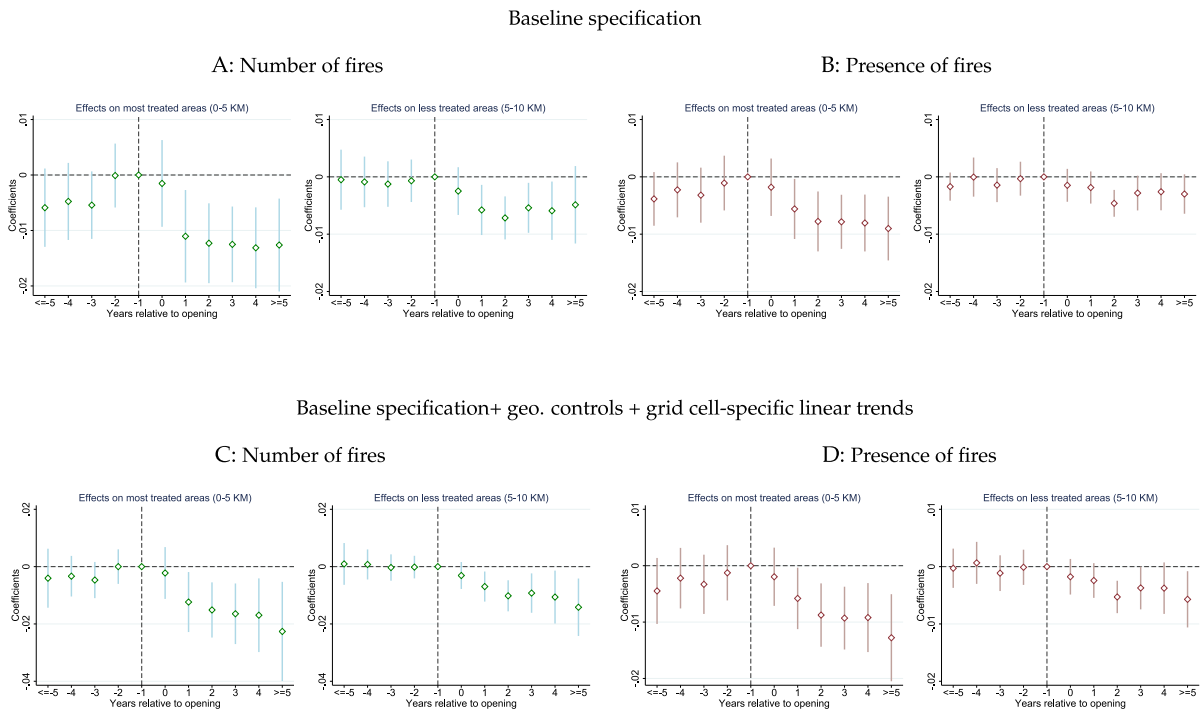


Fig. 4. Biomass power plants: Dynamic effect on agricultural fires.
Notes: The unit of observation is 1 km × 1 km grid cells. This figure checks the dynamic effect of biomass power plants on agricultural fires by conducting an event study plot. *Presence of fires* is an indicator for observing at least one fire. The year before a plant opening is omitted as the reference group. Panel A and B only control for grid cell fixed effects and plant × year fixed effects. Panel C and D further include grid cell-specific linear trends and a set of geographic controls interacted with year fixed effects. The diamonds denote coefficients and bars denote the associated 95% confidence intervals, where standard errors used to construct the confidence intervals are clustered at the plant level.

Table 2
The effect of biomass power plants on agricultural fires: Robustness.

Dep. var. =	(1) Treatment intensity # fires	(2) Treatment intensity # fires > 0	(3) Controlling for spillovers # fires	(4) Controlling for spillovers # fires > 0	(5) Controlling for city shocks # fires	(6) Controlling for city shocks # fires > 0	(7) Controlling for 2013 air quality monitoring # fires	(8) Controlling for 2013 air quality monitoring # fires > 0
Distance to the plant × Post	0.0013***	0.0007***						
	(0.0003)	(0.0002)						
<5 km of the plant × Post			−0.0132***	−0.0065***	−0.0122***	−0.0064***	−0.0127***	−0.0065***
			(0.0033)	(0.0017)	(0.0030)	(0.0015)	(0.0028)	(0.0015)
5–10 km of the plant × Post			−0.0072***	−0.0028***	−0.0067***	−0.0027***	−0.0068***	−0.0027***
			(0.0021)	(0.0011)	(0.0018)	(0.0010)	(0.0018)	(0.0009)
Distance to neighbor plant × Year FE	No	No	Yes	Yes	No	No	No	No
City × Year FE	No	No	No	No	Yes	Yes	No	No
Dist. to city center × Post 2013	No	No	No	No	No	No	Yes	Yes
Grid cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Plant × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geo. controls × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grid cell trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster level	Plant	Plant	Plant	Plant	Plant	Plant	Plant	Plant
Observations	1,986,576	1,986,576	1,611,456	1,611,456	1,986,560	1,986,560	1,966,912	1,966,912
R-squared	0.5196	0.2378	0.5341	0.2430	0.5203	0.2392	0.5196	0.2377
Mean of dep. var.	0.0220	0.0174	0.0242	0.0190	0.0220	0.0174	0.0222	0.0176

Notes: The unit of observation is 1 km × 1 km grid cells. This table shows the robustness of the baseline results. *<5* km *of the plant* and *5–10* km *of the plant* are indicators for grid cells within 5 km and 5–10 km of the plant, respectively. *Post* is an indicator for years after the plant opening. Geographic controls include longitude, latitude, slope, ruggedness, elevation, distance to rivers, and number of rivers. Columns (1)–(2) use a continuous treatment intensity where areas closer to a plant are assumed to have greater treatment. Columns (3)–(4) control for the spillover effects from neighbor plants. Columns (5)–(6) control for shocks at the city level by including city × year fixed effects. Columns (7)–(8) consider the large-scale air quality monitoring program launched in 2013. Standard errors in parentheses are clustered at the plant level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

where $Distance_{ij}$ denotes the distance from the centroid of grid cell i to plant j. The coefficient of interest is β, which is expected to be positive. Columns (1)–(2) of Table 2 confirm such conjecture and show that moving 1 km closer to the plant reduces the number of fires by 0.0013 and the probability that fires occur by 0.0007, accounting for 6% and 4% of the sample means, respectively. Fig. 5 further conducts an event study plot and shows no diverging pre-trends across different areas.

Spillover effects Results are robust to controlling for spatial spillovers. The baseline results focus on areas within 15 km of a plant. However, a plant may also collect crop residues farther away. If so, the treatment indicators could be contaminated in the sense that they could also capture the effects of other plants. As shown in columns (3)–(4) of Table 2, after adding the interaction between the distance from the centroid of each grid cell to the nearest neighbor plant and year fixed

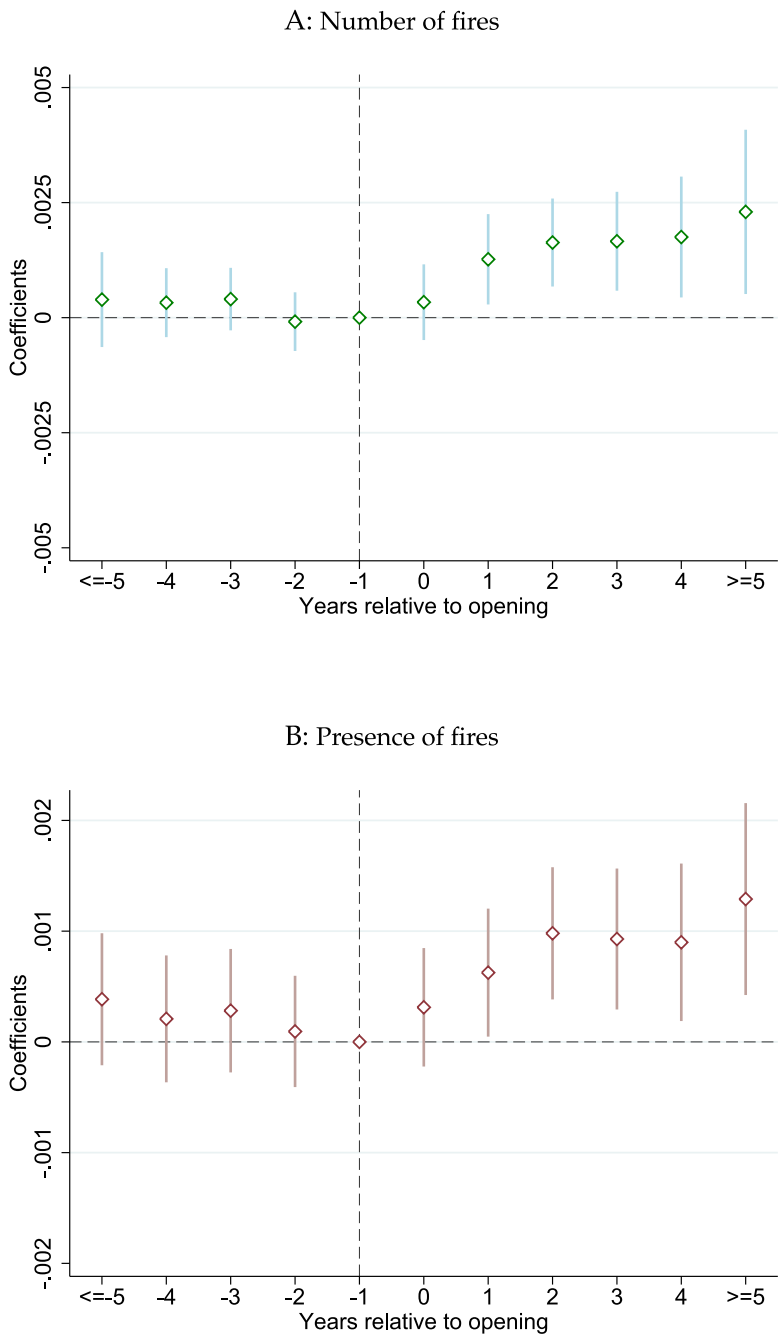


Fig. 5. Biomass power plants: Dynamic effect using continuous treatment intensity.
Notes: The unit of observation is 1 km × 1 km grid cells. This figure checks the dynamic effect of biomass power plants on agricultural fires by conducting an event study plot with continuous treatment intensity, which is the distance to the plant. *Presence of fires* is an indicator for observing at least one fire. The year before a plant opening is omitted as the reference group. Both Panel A and B control for grid cell fixed effects, plant × year fixed effects, grid cell-specific linear trends and a set of geographic controls interacted with year fixed effects. The diamonds denote coefficients and bars denote the associated 95% confidence intervals, where standard errors used to construct the confidence intervals are clustered at the plant level.

effects, the results barely change, suggesting that the spillover effects from other plants are negligible.

Confounding policies Results are not driven by confounding policies. In the sample, for about 48% of the plants, the 15 km buffers span more than one city. The treatment indicators could therefore capture differential city-level policies or economic conditions. To assess to what extent the results are driven by such factors, I add city × year fixed effects in columns (5)–(6) of Table 2 to flexibly control for all city-level factors. The results are not affected. I also add the interaction between the distance from the centroid of each grid cell to the city center and a post-2013 indicator to capture the large-scale air quality monitoring program launched in 2013. The program mainly aims at improving air quality in the urban areas. If the distance to a plant is correlated with the distance to the city center, then the results could be contaminated by such program. The results are unaffected as shown in columns (7)–(8) of Table 2. In Appendix G, I further examine five other major regulations in earlier years and the results stay virtually unchanged.

Persistence of fires Results are stable when I include a one-period lag of fires to control for the persistence of fires over time. Current burning could depend on past behavior as burning crop residues is a long-lasting tradition in China. Although including lagged dependent

Table 3
The effect of biomass power plants on agricultural fires: Robustness (Cont.)

Dep. var. =	(1) Controlling for persistence of fires # fires	(2) # fires > 0	(3) Clustering at: 2° × 2° # fires	(4) # fires > 0	(5) Conley standard errors # fires	(6) # fires > 0	(7) Placebo: fires in non-cropland # fires	(8) # fires > 0
<5 km of the plant × Post	−0.0124***	−0.0065***	−0.0127***	−0.0065***	−0.0127***	−0.0065***	−0.0048	−0.0012
	(0.0025)	(0.0016)	(0.0032)	(0.0014)	(0.0025)	(0.0015)	(0.0037)	(0.0011)
5–10 km of the plant × Post	−0.0059***	−0.0026***	−0.0068***	−0.0027***	−0.0068***	−0.0027***	0.0028	−0.0005
	(0.0015)	(0.0010)	(0.0018)	(0.0010)	(0.0016)	(0.0009)	(0.0046)	(0.0006)
Lagged dep. var. included	Yes	Yes	No	No	No	No	No	No
Grid cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Plant × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geo. controls × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grid cell trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster level	Plant	Plant	2° × 2°	2° × 2°	–	–	Plant	Plant
Observations	1,862,415	1,862,415	1,986,576	1,986,576	1,986,576	1,986,576	1,986,576	1,986,576
R-squared	0.5551	0.2560	0.5196	0.2378	0.5196	0.2378	0.8725	0.3193
Mean of dep. var.	0.0232	0.0183	0.0220	0.0174	0.0220	0.0174	0.0164	0.0071

Notes: The unit of observation is 1 km × 1 km grid cells. This table shows the robustness of the baseline results. *<5 km of the plant* and *5–10 km of the plant* are indicators for grid cells within 5 km and 5–10 km of the plant, respectively. *Post* is an indicator for years after the plant opening. Geographic controls include longitude, latitude, slope, ruggedness, elevation, distance to rivers, and number of rivers. Columns (1)–(2) control for the persistence of fires by including a one-period lag of fires. Columns (3)–(4) cluster the standard errors at the 2° × 2° level. Columns (5)–(6) adopt the Conley standard errors, assuming spatial correlation for observations within a 200 km radius and temporal correlation across 5 years. Columns (7)–(8) conduct a placebo test by replacing agricultural fires with non-agricultural fires. Standard errors in parentheses are clustered at the plant level in columns (1)–(2) and (7)–(8). * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

variable suffers from the well-known Nickell bias (Nickell, 1981), the results are almost unaffected as shown in columns (1)–(2) of Table 3.

Spatial correlation Results are not affected by controlling for spatial correlation. Given that both agricultural fires and biomass power plants are clustered in grain-producing areas, it is important to account for such spatial correlation. I first cluster the standard errors at the 2° × 2° level in columns (3)–(4) of Table 3. As one degree of longitude or latitude equals about 111 km at the equator, the clustering unit is large enough to capture most of the correlation. The standard errors show little change. I then allow the standard errors to be arbitrarily correlated across spatially adjacent observations using the Conley standard errors. Specifically, I assume spatial correlation for observations within a 200 km radius and temporal correlation across 5 years. The standard errors in columns (5)–(6) of Table 3 barely change.

Placebo tests The first placebo test is to replace the dependent variable with non-agricultural fires in columns (7)–(8) of Table 3. As expected, the coefficients are insignificant. I also conduct an event study plot for non-agricultural fires. The results are visualized in Fig. 6. Reassuringly, there is neither diverging trends before the opening nor a trend break after the opening. The second placebo test is to randomly permutate the timing of plant openings, holding the distribution of plant openings across years the same as the true distribution. I repeat this exercise 1000 times and plot the distribution of coefficient estimates in Appendix Figure A3. The distribution of placebo coefficients is not centered at zero since the placebo openings could to some extent coincide with the true openings. However, the true coefficients, denoted by the red lines, are far smaller than the placebo coefficients. These two placebo tests further justify the validity of the DID strategy.

4.5. Mechanisms

In this section, I first show that the previous findings are most likely consistent with an economic incentive-based explanation and then rule out several alternative explanations, namely, structural transformation, migration, and enhanced inspection.[20]

Economic Incentives A natural explanation of the findings is that the arrival of a plant increases the economic incentives to preserve crop residues, or equivalently increases the opportunity cost of burning crop residues. Specifically, before the arrival of a plant, the opportunity cost of burning is close to zero, as crop residues have basically no use for farmers. The opportunity cost increases sharply after the arrival of a plant since farmers are paid for their crop residues if sold to the plant. In Appendix F.1, I provide some qualitative evidence that corroborates such mechanism. In particular, I show that collecting crop residues are generally doable for farmers and the revenue from this also accounts for a non-trivial proportion of total value of agricultural production. Furthermore, farmers are also aware of the benefits of selling crop residues and view the revenue as attractive.

I test such mechanism more formally by looking at areas where economic incentives are lower. In such areas, one should see a smaller effect. In columns (1)–(4) of Table 4, I use the slope and ruggedness of each grid cell to measure economic incentives, as in areas with large slope or ruggedness it would be costly to collect crop residues. I do see a smaller effect in grid cells with larger slope or ruggedness, although it is only statistically significant for the most treated areas. In columns (5)–(6), I further measure the difficulty to collect crop residues using the number of fragments created by rivers in a grid cell. The intuition is simple: if a grid cell is cut into a larger number of disconnected areas, then collecting the crop residues in that grid cell becomes more costly as one need to cross the rivers for more times. This idea is similar to Cutler and Glaeser (1997) and Bai and Jia (2016) who use rivers to construct measures of administrative costs in an area.[21] I find a smaller effect for areas that are more fragmented, and the effect is statistically significant for both the most treated areas and the less treated areas. Therefore, the results in this table are supportive of an economic incentive mechanism. I further exploit household survey data, although less granular, in Appendix F.2 to show that farmers' income indeed increases after a plant opening. Taken together, both the qualitative and quantitative evidence point to a mechanism of economic incentives behind the reduction in fires.

Structural Transformation The reduction in agricultural fires after a plant opening could be simply caused by a reduction in farming. This could happen if the plant directly employs farmers nearby, or if

[20] Appendix F.6 further rules out an additional mechanism: potential fire bans under transmission towers.

[21] Slightly different from Cutler and Glaeser (1997) and Bai and Jia (2016), I do not directly use the number of rivers as in this setting the number of rivers might affect agricultural production. I instead include the number of rivers (interacted with year fixed effects) as control variables.

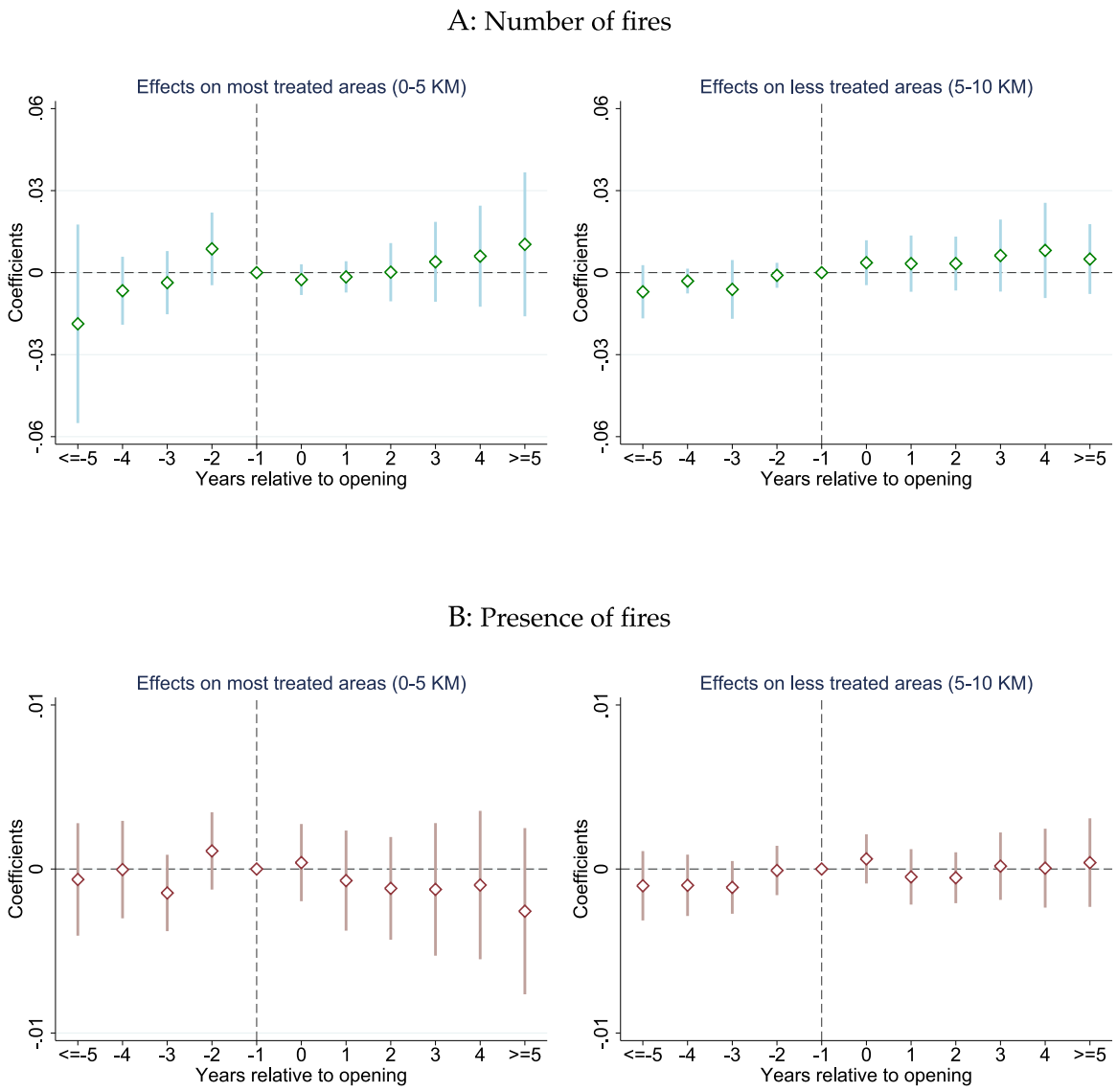


Fig. 6. Biomass power plants: Dynamic effect on non-agricultural fires.
Notes: The unit of observation is 1 km × 1 km grid cells. This figure checks the dynamic effect of biomass power plants on non-agricultural fires by conducting an event study plot. *Presence of fires* is an indicator for observing at least one fire. The year before a plant opening is omitted as the reference group. Both Panel A and B control for grid cell fixed effects, plant × year fixed effects, grid cell-specific linear trends, and a set of geographic controls interacted with year fixed effects. The diamonds denote coefficients and bars denote the associated 95% confidence intervals, where standard errors used to construct the confidence intervals are clustered at the plant level.

the plant stimulates the rise of other firms, which indirectly affect the economic structure (Dell and Olken, 2020).

To test such conjecture, I first check the effects of plant openings on land usage in columns (1)–(2) of Table 5. The coefficients are small and statistically insignificant, indicating little change in cropland or settlement land. I then check the effects of plant openings on industrial dynamics in the remaining columns of Table 5. In columns (3)–(4), I test the effect on the number of industrial firms and employees in each grid cell based on the ASIF data. Similarly, in columns (5)–(6), I test the effect on real capital and output.[22] Again, the coefficients are small and statistically insignificant, implying that there is little change in the industrial sector.[23] As the ASIF data do not capture the dynamics of small firms, I supplement such finding using additional data that cover the entire population of industrial firms in Appendix F.3 and the results are similar. Taken together, these results suggest that my baseline findings are unlikely to be driven by changes in farming caused by plant openings.

Relatedly, even if there is no substantial change in land usage, cropland and agricultural production near the plants may be different

[22] Output is measured by value added. Capital is the capital stock calculated following the procedure in Brandt et al. (2012). Both variables are then converted into real values using industry-level price indices.

[23] Several reasons could explain why the plants have little impact on local economic activity. First, the electricity generated is tapped into distribution grids and electrification rate in China was already close to 100% many years ago (see https://data.worldbank.org/indicator/EG.ELC.ACCS.ZS?locations=CN), implying that the arrival of the plants have minimal effect on local economic activity through electrification. Second, given that most plants utilize technologies and machines imported from abroad, it is unlikely that the plants will significantly stimulate related industries locally. From the perspective of economics of density, agglomeration effect of the plants, if any, is also limited by the sparse distribution of industrial activities in the rural areas. Finally, the plants hire relative few formal employees, which is unlikely to generate significant economic impact through affecting the local labor market.

Table 4
Biomass power plants: Testing the economic incentive mechanism.

Dep. var. =	(1) # Fires	(2) # Fires > 0	(3) # Fires	(4) # Fires > 0	(5) # Fires	(6) # Fires > 0
<5 km of the plant × Post	−0.0138***	−0.0073***	−0.0139***	−0.0073***	−0.0225***	−0.0147***
	(0.0032)	(0.0017)	(0.0032)	(0.0017)	(0.0071)	(0.0036)
5–10 km of the plant × Post	−0.0074***	−0.0029***	−0.0074***	−0.0029***	−0.0183***	−0.0132***
	(0.0020)	(0.0011)	(0.0021)	(0.0011)	(0.0045)	(0.0031)
Slope × <5 km of the plant × Post	0.0009**	0.0007**				
	(0.0005)	(0.0003)				
Slope × 5–10 km of the plant × Post	0.0004	0.0002				
	(0.0003)	(0.0002)				
Ruggedness × <5 km of the plant × Post			0.0000**	0.0000**		
			(0.0000)	(0.0000)		
Ruggedness × 5–10 km of the plant × Post			0.0000	0.0000		
			(0.0000)	(0.0000)		
# Fragments created by rivers × Post					−0.0008	−0.0011
					(0.0032)	(0.0018)
# Fragments created by rivers × <5 km of the plant × Post					0.0088*	0.0074***
					(0.0048)	(0.0026)
# Fragments created by rivers × 5–10 km of the plant × Post					0.0104***	0.0096***
					(0.0034)	(0.0024)
Grid cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Plant × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo. controls × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Grid cell trends	Yes	Yes	Yes	Yes	Yes	Yes
Cluster level	Plant	Plant	Plant	Plant	Plant	Plant
Observations	1,986,576	1,986,576	1,986,576	1,986,576	1,986,576	1,986,576
R-squared	0.5196	0.2378	0.5196	0.2378	0.5196	0.2378
Mean of dep. var.	0.0220	0.0174	0.0220	0.0174	0.0220	0.0174

Notes: The unit of observation is 1 km × 1 km grid cells. This table shows that the reduction of agricultural fires after a plant opening is consistent with an economic incentive explanation. *<5* km *of the plant* and *5–10* km *of the plant* are indicators for grid cells within 5 km and 5–10 km of the plant, respectively. *Post* is an indicator for years after the plant opening. Geographic controls include longitude, latitude, slope, ruggedness, elevation, distance to rivers, and number of rivers. Columns (1)–(2) and (3)–(4) show smaller effects in areas with greater difficulty in collecting crop residues, where difficulty is measured by slope and ruggedness, respectively. Columns (5)–(6) also show smaller effects in areas with greater difficulty in collecting crop residues, where difficulty is measured by the number of disconnected fragments created by rivers in a grid cell. Standard errors in parentheses are clustered at the plant level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

Table 5
Biomass power plants: Testing the structural transformation mechanism.

Dep. var. =	(1) Land usage	(2)	(3) Industrial sector dynamics	(4)	(5)	(6)
	% Crop	% Settlement	# Firms	# Employees	log(Capital+1)	log(Output+1)
<5 km of the plant × Post	0.0006	−0.0009	0.0555	1.9697	0.0585	0.0274
	(0.0010)	(0.0013)	(0.0820)	(14.7630)	(0.0453)	(0.0734)
5–10 km of the plant × Post	−0.0001	−0.0005	−0.0143	−5.5121	0.0522	0.0655
	(0.0006)	(0.0007)	(0.0734)	(10.6314)	(0.0376)	(0.0444)
Grid cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Plant × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo. controls × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Grid cell trends	Yes	Yes	Yes	Yes	Yes	Yes
Cluster level	Plant	Plant	Plant	Plant	Plant	Plant
Observations	1,986,576	1,986,576	869,127	869,127	869,127	744,966
R-squared	0.9943	0.9810	0.8010	0.8452	0.8676	0.8859
Mean of dep. var.	0.734	0.0761	0.203	42.32	0.474	0.548

Notes: The unit of observation is 1 km × 1 km grid cells. This table shows that the reduction of agricultural fires after a plant opening is not caused by changes of economic structure. *<5* km *of the plant* and *5–10* km *of the plant* are indicators for grid cells within 5 km and 5–10 km of the plant, respectively. *Post* is an indicator for years after the plant opening. Geographic controls include longitude, latitude, slope, ruggedness, elevation, distance to rivers, and number of rivers. Standard errors in parentheses are clustered at the plant level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

due to negative externalities, such as air pollution and noise, imposed by the plants. In Appendix F.4, I show through a case study that a typical plant imposes negligible negative environmental externalities on nearby cropland, and that the results are robust to dropping areas close to a plant. Furthermore, farmers may also adapt to the plant by adjusting along various dimensions of agricultural production, such as crop choice, harvest techniques, labor usage, timing of harvest/planting or even productivity. These changes may not only affect crop residues and fires, but also be interesting in themselves. In Appendix F.5, I show that there is little change in crop choice and agricultural productivity. For harvest techniques, labor usage and timing of harvest/planting, I cannot test them directly due to lack of granular data. However, these changes should be largely reflected in the change of the value of agricultural production, which is not the case as shown in Appendix F.5.

Migration Another explanation for the findings is that farmers migrate away from a plant to avoid possible air pollution caused by the plant (Chen et al., 2017b; Barwick et al., 2021). In this case, the cropland may be less utilized and generate fewer agricultural fires. To clarify, migration typically refers to long-term or permanent movement of residence. This form of migration is largely restricted by the *hukou* system in China. However, short-term change of residence

Table 6
Biomass power plants: Testing the migration mechanism.

Dep. var. =	(1)	(2)	(3)	(4)	(5)	(6)
	Downwind is measured as: % days downwind of the plant in a year		1 (% days downwind > plant-year median)		Road access: 1(within 1 km of railways or major roads)	
	#Fires	#Fires > 0	#Fires	#Fires > 0	#Fires	#Fires > 0
<5 km of the plant × Post	−0.0146***	−0.0067***	−0.0139***	−0.0066***	−0.0107***	−0.0066***
	(0.0048)	(0.0024)	(0.0039)	(0.0017)	(0.0025)	(0.0016)
5–10 km of the plant × Post	−0.0116***	−0.0036**	−0.0084***	−0.0028**	−0.0052***	−0.0021**
	(0.0034)	(0.0018)	(0.0024)	(0.0012)	(0.0019)	(0.0010)
Downwind	−0.0006	−0.0017	0.0009	0.0003		
	(0.0045)	(0.0030)	(0.0013)	(0.0008)		
Downwind × Post	−0.0083	0.0021	−0.0016	0.0005		
	(0.0104)	(0.0044)	(0.0025)	(0.0012)		
Downwind × <5 km of the plant	−0.0069	−0.0023	−0.0021	−0.0007		
	(0.0083)	(0.0051)	(0.0025)	(0.0015)		
Downwind × 5–10 km of the plant	−0.0028	−0.0011	−0.0024	−0.0012		
	(0.0060)	(0.0040)	(0.0016)	(0.0011)		
Downwind × <5 km of the plant × Post	0.0071	0.0006	0.0023	0.0001		
	(0.0123)	(0.0068)	(0.0037)	(0.0019)		
Downwind × 5–10 km of the plant × Post	0.0196*	0.0034	0.0032	0.0002		
	(0.0100)	(0.0058)	(0.0025)	(0.0016)		
Road access × Post					0.0108	0.0025
					(0.0067)	(0.0018)
Road access × <5 km of the plant × Post					−0.0114	−0.0002
					(0.0090)	(0.0033)
Road access × 5–10 km of the plant × Post					−0.0095	−0.0031
					(0.0060)	(0.0027)
Grid cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Plant × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo. controls × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Grid cell trends	Yes	Yes	Yes	Yes	Yes	Yes
Cluster level	Plant	Plant	Plant	Plant	Plant	Plant
Observations	1,963,011	1,963,011	1,963,011	1,963,011	1,986,576	1,986,576
R-squared	0.5215	0.2387	0.5215	0.2387	0.5196	0.2378
Mean of dep. var.	0.0219	0.0173	0.0219	0.0173	0.0220	0.0174

Notes: The unit of observation is 1 km × 1 km grid cells. This table shows that the reduction of agricultural fires after a plant opening is not caused by potential pollution-induced migration. *<5 km of the plant* and *5–10 km of the plant* are indicators for grid cells within 5 km and 5–10 km of the plant, respectively. *Post* is an indicator for years after the plant opening. Geographic controls include longitude, latitude, slope, ruggedness, elevation, distance to rivers, and number of rivers. Columns (1)–(4) show that the effects are not larger in downwind areas of a plant that are perceived to suffer more from potential pollution. Columns (5)–(6) show no differential effects in areas with greater ease of migration, proxied by access to transportation networks. Standard errors in parentheses are clustered at the plant level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

is still possible because farmers may migrate temporarily to seek for seasonal work in big cities. Conceptually, pollution-induced migration is unlikely to happen since the biomass power plants use a cleaner technology to process the crop residues than farmers. Moreover, as pointed out by Chen et al. (2017b), migration usually happens at a longer horizon such as a five-year period. As I focus on annual changes, the results are unlikely to be affected by migration.

Empirically, I test such conjecture in the following ways. I first test whether the effects are larger in areas downwind of the plant, since such areas are proved to suffer more from possible pollution (Rangel and Vogl, 2019; Anderson, 2020). The procedure to determine downwind areas can be found in Section 3.2 and is also illustrated in Appendix Figure A1. In columns (1)–(2) of Table 6, I use a continuous variable that is the percentage of days downwind of the plant in a year to measure a grid cell's exposure to pollution. In columns (3)–(4), I further create an indicator for grid cells where the percentage of days downwind of the plant in a year is greater than the median percentage for that plant and year. The coefficients on the triple interaction terms in columns (1)–(4) are either statistically insignificant or have a positive sign, implying that the findings are unlikely to be driven by pollution-induced migration. In columns (5)–(6), I test whether the effects are larger in grid cells where it would be easier to migrate. I define such grid cells as those whose centroids are within 1 km of major transportation networks. I use the major roads and railways in 2000 due to data limitation. I cannot find a larger effect in areas with greater ease of migration. Therefore, migration cannot explain the findings.

Enhanced Inspection The arrival of a plant could signal local officials' attention paid to agricultural fires. Therefore, the findings could be driven by enhanced inspection by local officials that occurs concurrently with the plant opening.

In columns (1)–(2) of Table 7, I test whether the effects are larger in grid cells closer to the city center, as monitoring costs are lower in such areas (Huang et al., 2017). I measure the proximity to city center using the distance from the centroid of each grid cell to the center of the city where the plant is located. I cannot find a larger effect in such areas. In columns (3)–(4), I test whether the effects are weaker in grid cells that are located in a city different from the city where the plant is located, as local officials have little incentive to control pollution outside their jurisdictions (Lipscomb and Mobarak, 2016). I fail to find any differential effects. In columns (5)–(6), I test whether the effects are stronger in grid cells that are located in a city where the leader has stronger promotion incentives, as pollution control is considered by upper-level governments for promotion. I define high promotion incentives if the city party secretary is younger than 57, as the chance for promotion is little for secretaries older than 57 (He et al., 2020b).[24] I cannot find a larger effect for areas where the secretaries have stronger promotion incentives. Hence, the drop in agricultural fires cannot be explained by local officials' actions.

[24] Appendix Table A3 shows that the results are robust if I define promotion incentives using the mayor's age.

Table 7
Biomass power plants: Testing the enhanced inspection mechanism.

Dep. var. =	(1)	(2)	(3) Other city: 1(grid cell and the plant are located in different cities)	(4)	(5) High incentive: 1(city party secretary is younger than 57)	(6)
	# Fires	# Fires > 0	# Fires	# Fires > 0	# Fires	# Fires > 0
<5 km of the plant × Post	−0.0145***	−0.0072***	−0.0059***	−0.0120***	−0.0102***	−0.0064***
	(0.0049)	(0.0022)	(0.0014)	(0.0030)	(0.0037)	(0.0024)
5–10 km of the plant × Post	−0.0099***	−0.0044***	−0.0024**	−0.0068***	−0.0055**	−0.0021*
	(0.0035)	(0.0015)	(0.0010)	(0.0019)	(0.0026)	(0.0013)
Distance to city center × Post	−0.0001	−0.0000				
	(0.0001)	(0.0000)				
Distance to city center × <5 km of the plant × Post	0.0000	0.0000				
	(0.0001)	(0.0000)				
Distance to city center × 5–10 km of the plant × Post	0.0001	0.0000				
	(0.0000)	(0.0000)				
Other city × Post			0.0029	0.0043		
			(0.0031)	(0.0066)		
Other city × <5 km of the plant × Post			−0.0085	−0.0104		
			(0.0100)	(0.0131)		
Other city × 5–10 km of the plant × Post			−0.0022	0.0027		
			(0.0060)	(0.0074)		
High incentive					0.0041	0.0034
					(0.0031)	(0.0021)
High incentive × Post					−0.0016	−0.0031
					(0.0046)	(0.0029)
High incentive × <5 km of the plant					0.0005	−0.0002
					(0.0022)	(0.0014)
High incentive × 5–10 km of the plant					0.0027	0.0013
					(0.0025)	(0.0010)
High incentive × <5 km of the plant × Post					−0.0034	0.0004
					(0.0034)	(0.0022)
High incentive × 5–10 km of the plant × Post					−0.0018	−0.0005
					(0.0029)	(0.0014)
Grid cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Plant × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo. controls × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Grid cell trends	Yes	Yes	Yes	Yes	Yes	Yes
Cluster level	Plant	Plant	Plant	Plant	Plant	Plant
Observations	1,966,912	1,966,912	1,986,576	1,986,576	1,858,853	1,858,853
R-squared	0.5196	0.2377	0.2378	0.5196	0.5248	0.2414
Mean of dep. var.	0.0220	0.0174	0.0174	0.0220	0.0220	0.0174

Notes: The unit of observation is 1 km × 1 km grid cells. This table shows that the reduction of agricultural fires after a plant opening is not caused by enhanced inspection by local officials following the arrival of the plant. *<5 km of the plant* and *5–10 km of the plant* are indicators for grid cells within 5 km and 5–10 km of the plant, respectively. *Post* is an indicator for years after the plant opening. Geographic controls include longitude, latitude, slope, ruggedness, elevation, distance to rivers, and number of rivers. Columns (1)–(2) show no differential effects in areas with greater penetration of local officials (areas closer to city center). Columns (3)–(4) show no differential effects in areas outside the city where the plant is located. Columns (5)–(6) show no differential effects in areas where the city party secretary has greater promotion incentives (younger than 57). Standard errors in parentheses are clustered at the plant level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

5. The effects of no burning zones

5.1. Model specification

Since exposure to a no burning zone is a deterministic and discontinuous function of the distance from each area to the 15 km-radius circle around the airport, I use a spatial regression discontinuity design to estimate the effects of no burning zones. Following Hahn et al. (2001), I specify a local linear regression:

$$Y_{isjt} = \delta_{sj} + \beta_1 Inside_{ij} + \beta_2 Distance_{ij} + \beta_3 Inside_{ij} \times Distance_{ij} + \varepsilon_{isjt} \quad (7)$$

$$s.t. \ -h \le Distance_{ij} \le h$$

where the unit of observation is 1 km × 1 km grid cells. i denotes grid cells, s denotes boundary segments, j denotes airports, and t denotes years. $Inside_{ij}$ is an indicator equal to 1 if the centroid of grid cell i is inside the no burning zone around airport j and 0 otherwise. The sample used here only includes active airports, so the treatment variable $Inside_{ij}$ is time-invariant. I also utilize the panel structure of the data by including the years before an airport is active and estimating a difference-in-discontinuities specification in the robustness checks. $Distance_{ij}$ is the running variable, measuring the distance from the centroid of grid cell i to the no burning zone boundary around airport j (negative for observations outside the boundary). I include the interaction between $Inside_{ij}$ and $Distance_{ij}$ to allow for the coefficients on $Distance_{ij}$ to be different for each side of the boundary. The coefficient of interest is β_1, which captures the jump of outcomes at the boundary. To ensure the areas just outside the boundary are valid counterfactuals for areas just inside the boundary, I split the boundary into 10 segments of equal length and assign each grid cell to the nearest segment. I then include airport × segment fixed effects δ_{sj}, which ensures that I am comparing grid cells within a narrowly defined geographic area. Due to spatial adjacency, grid cells along the same segment should be ex-ante similar.

For most of the analysis, I report the results using three fixed bandwidths: 2 km, 4 km, 6 km, and also the optimal bandwidth calculated following the procedure proposed by Imbens and Kalyanaraman (2012).[25] The optimal bandwidths for different outcomes and

[25] For specifications with fixed effects and predetermined covariates, the optimal bandwidth is selected based on the residuals from regressing the

specifications generally range from 2 km to 6 km with a mean value of about 4 km. I also check the robustness of the baseline results to a broader set of bandwidths ranging from 2 km to 10 km. I weight observations within the bandwidth using a uniform kernel for convenience and check robustness to a triangular kernel, as in practice the choice of kernels has little impact (Lee and Lemieux, 2010). I cluster the standard errors at the airport level to account for spatial and temporal correlation (Cameron and Miller, 2015). I also show robustness to clustering at the 2° × 2° level and accounting for arbitrary correlation across spatially adjacent observations (Conley, 1999; Hsiang, 2010).

5.2. Identification concerns

The key identification assumption of this RD design is that all relevant preexisting factors change very smoothly around a no burning zone boundary. This assumption is needed to ensure areas just outside the boundary are an appropriate counterfactual for those located just inside the boundary. Conceptually, this assumption is plausibly satisfied since the boundary is purely artificially drawn and does not overlap with other administrative boundaries. Nevertheless, I check for possible jumps at the boundary for a set of preexisting covariates, including geography, location, land usage, and economic structure.[26]

Another threat to the identification assumption is that there could be selective sorting of farmers around the boundary.[27] As mentioned before, long-term or permanent change of residence is hard for farmers due to the *hukou* system, largely alleviating the selective sorting concern. Nevertheless, selective sorting may still occur in other forms. First, cropland inside the boundary may be converted to commercial land, either by farmers to avoid regulation or by local governments to avoid accountability for agricultural fires. Conceptually, this is unlikely to happen since cropland is strictly protected by the central government.[28] For farmers, it is illegal to change the usage of cropland. For local governments, they need to get the approval from the central government for changes. Empirically, I check this possibility by looking at the change of land usage ex post. Second, local governments may implicitly encourage industrial firms to locate inside the boundary to transform the economic structure. I test this possibility by looking at the number and size of industrial firms using the ASIF data. Third, farmers inside the boundary may farm outside the boundary to avoid regulation. Conceptually, this is unlikely to happen since cropland is owned collectively and each farmer only owns the use rights of a small piece of land near their house. To get cropland outside, one needs to negotiate with farmers there and get the approval of relevant authorities. Moreover, it would also be physically costly to farm in areas far away. Empirically, I adopt a "donut hole" specification by excluding areas close to the boundary (Barreca et al., 2011, 2016), since farmers living in these areas are likely to have strong incentive to cross the boundary. Fourth, farmers inside the boundary may move out temporarily to seek for seasonal work in big cities. I test this possibility by looking at areas with greater ease of migration, which are areas with access to major transportation networks. Finally, selective sorting may also occur at the airport level. For the existing airport, it is nearly impossible to move since it is too costly. However, local governments may choose to locate new airports in some areas with very few agricultural fires to avoid accountability. I check this possibility by looking at airports established before the launch of the regulation in 1999.

dependent variable on these fixed effects and predetermined covariates (Lee and Lemieux, 2010).

[26] I do not compare climate variables such as temperature and precipitation because high resolution data are not available. Moreover, across the grid cells around an airport, such variables should not vary much.

[27] Selective sorting could happen as an outcome of the discontinuous policy, but then the interpretation would be different.

[28] See http://www.npc.gov.cn/npc/c2168/200011/048bf4075bee4b4da938a7681d63b58f.shtml.

5.3. Results

Baseline RD results Table 8 reports the RD results that identify the effects of no burning zones on agricultural fires using three fixed bandwidths: 2 km, 4 km, and 6 km, and the optimal bandwidth. Columns (1)–(3) examine the number of fires. The coefficients are small compared to the sample mean, and statistically insignificant. Moreover, they remain quite stable as more granular fixed effects are included. The coefficients change slightly when using larger bandwidths, but are still negligible compared to the sample mean. Columns (4)–(6) examine the probability that any fires occur. The patterns are quite similar to those in columns (1)–(3). Across all specifications and bandwidths, the coefficients range from −0.0004 to −0.0001, which are economically small compared to the sample mean of 0.009 and are statistically insignificant. Overall, these results suggest that the regulation fails to reduce agricultural fires. In Appendix Figure A6, I report the RD results using different bandwidths ranging from 2 km to 10 km. The results are not sensitive to alternative bandwidths and are still small and insignificant.

Fig. 7 visualizes the RD results. The circles denote the average number of fires or probability that any fires occur within a distance bin, after partialling out the airport × segment fixed effects. The lines are fitted separately for grid cells on each side of the boundary, and the dashes are the associated 95% confidence intervals. In line with the RD estimates in Table 8, the jumps at the boundary are small. Moreover, the confidence intervals at the boundary are largely overlapped, implying that the difference in fires across the boundary is statistically indistinguishable from zero.

As discussed in Section 3, classical measurement errors in agricultural fires could decrease the precision of the estimates. However, as the economic magnitude is also close to zero, it is then hard to conclude that the zones really work. Even if the measurement errors are non-random due to farmers timing the burning, then this would only bias my findings against zero; as I do not find any reduction, this further strengthens my conclusion that the zones fail to reduce burning.

Preexisting balance I conduct a balance check for the preexisting covariates around the boundary, using the same specification as Eq. (7). Except for the distance to major transportation networks that is measured in 2000 due to data limitation,[29] all other time-varying variables are measured in 1998, one year before the launch of the regulation in 1999. The results are reported in Appendix Table A22, using three fixed bandwidths: 2 km, 4 km, and 6 km, and the optimal bandwidth. The coefficients are generally small, compared to the sample means, and statistically insignificant for most of the coefficients. A few covariates, such as the distance to city border, the share of settlement land, and the nighttime light intensity in 1998, show statistically significant jump at the boundary, but the magnitude of the jump is negligible compared to the sample mean. In Appendix Figure A4 and Appendix Figure A5, I show the RD plots for all these covariates using the optimal bandwidth. In line with the results in Appendix Table A22, there are no discernible jumps at the boundary.

Selective sorting I check for possible selective sorting in Table 9, still adopting the same specification as Eq. (7) and using three fixed bandwidths: 2 km, 4 km, and 6 km, and the optimal bandwidth. In columns (1)–(2), I find no evidence of differences in land usage, either for farming or for settlement, across the boundary. The coefficients are small and statistically insignificant, alleviating the concern that farmers or local governments may change land usage as a response to the regulation. In columns (3)–(6), I test the difference in economic structure, measured by the number and size of industrial firms, across

[29] The transportation networks in 2000 can also be viewed as predetermined as they were likely to be planned several years before 2000.

Table 8
The effect of no burning zones on agricultural fires.

Dep. var. =	(1) Number of fires	(2)	(3)	(4) Presence of fires	(5)	(6)
			Panel A: Bandwidth = 2 km			
Inside	0.0004	0.0006	0.0005	−0.0004	−0.0002	−0.0003
	(0.0008)	(0.0007)	(0.0007)	(0.0005)	(0.0005)	(0.0005)
Observations	914,902	914,902	914,902	914,902	914,902	914,902
Mean of dep. var.	0.0115	0.0115	0.0115	0.0097	0.0097	0.0097
			Panel B: Bandwidth = *4* km			
Inside	0.0000	0.0001	0.0001	−0.0002	−0.0001	−0.0001
	(0.0007)	(0.0007)	(0.0007)	(0.0004)	(0.0004)	(0.0004)
Observations	1,829,849	1,829,849	1,829,849	1,829,849	1,829,849	1,829,849
Mean of dep. var.	0.0110	0.0110	0.0110	0.0092	0.0092	0.0092
			Panel C: Bandwidth = *6* km			
Inside	−0.0004	−0.0004	−0.0004	−0.0002	−0.0001	−0.0001
	(0.0008)	(0.0008)	(0.0008)	(0.0004)	(0.0004)	(0.0004)
Observations	2,743,315	2,743,315	2,743,315	2,743,315	2,743,315	2,743,315
Mean of dep. var.	0.0109	0.0109	0.0109	0.0092	0.0092	0.0092
			Panel D: Optimal bandwidth (km)			
Inside	0.0007	0.0008	0.0004	−0.0001	−0.0001	−0.0001
	(0.0007)	(0.0007)	(0.0007)	(0.0004)	(0.0004)	(0.0004)
Observations	1,030,204	1,016,363	1,107,607	1,790,391	1,799,071	1,902,847
Mean of dep. var.	0.0114	0.0115	0.0114	0.0092	0.0092	0.0092
Bandwidth	2.2541	2.2222	2.4227	3.9146	3.9343	4.1610
RD polynomial	Linear	Linear	Linear	Linear	Linear	Linear
Airport FE	No	Yes	Yes	No	Yes	Yes
Airport × Segment FE	No	No	Yes	No	No	Yes
Cluster level	Airport	Airport	Airport	Airport	Airport	Airport

Notes: The unit of observation is 1 km × 1 km grid cells. The no burning zones are areas within 15 km of an airport. This table shows that no burning zones do not reduce agricultural fires, by conducting a spatial RD estimation around the no burning zone boundaries. *Presence of fires* is an indicator for observing at least one fire. *Inside* is an indicator for areas within the no burning zones. The optimal bandwidth in Panel D is selected using the procedure proposed by Imbens and Kalyanaraman (2012). Standard errors in parentheses are clustered at the airport level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

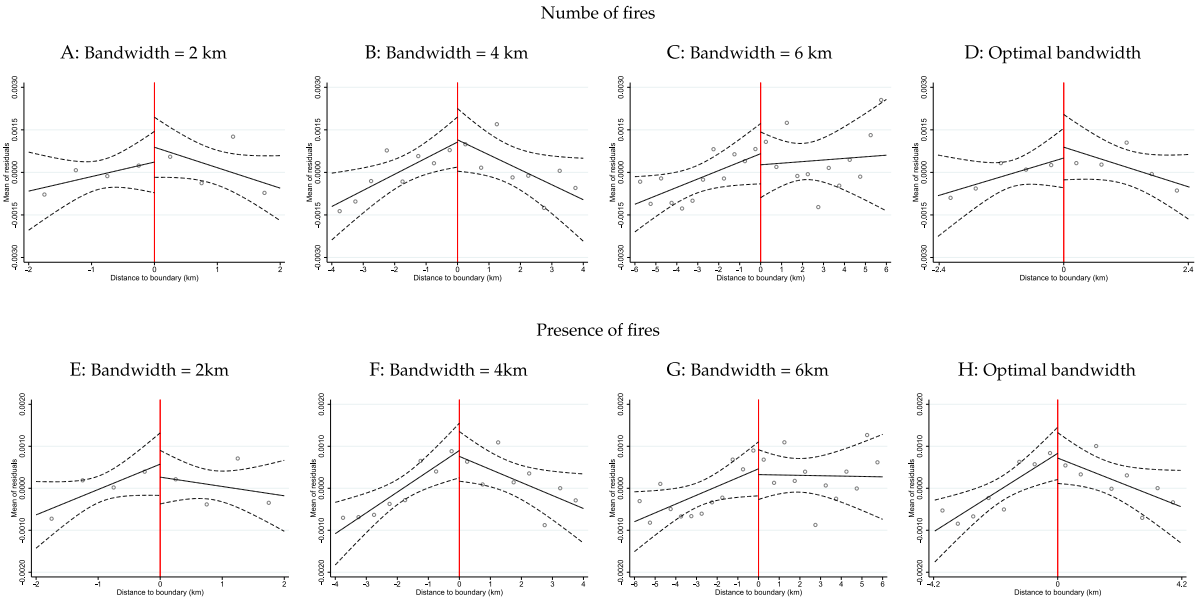


Fig. 7. No burning zones: Insignificant effect on agricultural fires - RD plot.
Notes: The unit of observation is 1 km × 1 km grid cells. The no burning zones are areas within 15 km of an airport. This figure shows that the no burning zones have little effect on agricultural fires. *Presence of fires* is an indicator for observing at least one fire. The circles denote the average number of fires or probability that any fires occur within a distance bin, after partialling out the airport × boundary segment fixed effects. The lines are fitted separately for grid cells on each side of the boundary, and the dashes are the associated 95% confidence intervals, where standard errors used to construct the confidence intervals are clustered at the airport level. The optimal bandwidth in Pane D and H are selected using the procedure proposed by Imbens and Kalyanaraman (2012).

Table 9
The effect of no burning zones on agricultural fires: Testing for sorting.

Dep. var. =	(1) Checking the land usage and economic structure		(3)	(4)	(5)	(6)	(7) Donut hole specification	(8)	(9) Ease of migration: Hard	(10) Easy	(11) Hard	(12) Easy	(13) Only airports built before 1999	(14)
	% Cropland	% Settlement land	# Firms	# Employees	log(1+capital)	log(1+output)	# Fires	# Fires > 0	# Fires	# Fires	# Fires > 0	# Fires > 0	# Fires	# Fires > 0
Panel A: Bandwidth = 2 km														
Inside	0.0012	−0.0009	0.0485**	9.1964*	−0.0121	−0.0136	0.0014	0.0006	0.0000	0.0033	−0.0006	0.0012	0.0005	−0.0003
	(0.0034)	(0.0016)	(0.0241)	(5.5490)	(0.0266)	(0.0316)	(0.0013)	(0.0007)	(0.0007)	(0.0027)	(0.0005)	(0.0014)	(0.0010)	(0.0006)
Observations	914,902	914,902	327,552	327,552	327,552	280,801	915,503	915,503	757,879	157,020	757,879	157,020	615,136	615,136
Mean of dep. var.	0.4670	0.0686	0.2330	48.6800	0.5220	0.5970	0.0111	0.0092	0.0106	0.0158	0.0090	0.0128	0.0126	0.0106
Panel B: Bandwidth = 4 km														
Inside	0.0002	0.0002	0.0189	2.5191	−0.0085	−0.0116	0.0001	0.0005	−0.0006	0.0035	−0.0004	0.0015	0.0004	0.0002
	(0.0039)	(0.0018)	(0.0225)	(5.1880)	(0.0249)	(0.0285)	(0.0011)	(0.0005)	(0.0008)	(0.0024)	(0.0005)	(0.0013)	(0.0009)	(0.0005)
Observations	1,829,849	1,829,849	655,155	655,155	655,155	561,664	1,828,918	1,828,918	1,514,760	315,088	1,514,760	315,088	1,230,096	1,230,096
Mean of dep. var.	0.4680	0.0686	0.2310	47.3100	0.5090	0.5830	0.0107	0.0090	0.0101	0.0151	0.0086	0.0121	0.0119	0.0100
Panel C: Bandwidth = 6 km														
Inside	−0.0001	0.0014	0.0047	1.8606	−0.0148	−0.0189	−0.0013	−0.0001	−0.0004	0.0013	−0.0002	0.0010	0.0003	0.0002
	(0.0041)	(0.0022)	(0.0223)	(4.3612)	(0.0238)	(0.0272)	(0.0013)	(0.0005)	(0.0008)	(0.0025)	(0.0005)	(0.0011)	(0.0009)	(0.0005)
Observations	2,743,315	2,743,315	982,154	982,154	982,154	842,002	2,742,818	2,742,818	2,270,713	472,600	2,270,713	472,600	1,844,016	1,844,016
Mean of dep. var.	0.4680	0.0679	0.2260	46.4000	0.5060	0.5790	0.0110	0.0091	0.0100	0.0155	0.0085	0.0123	0.0117	0.0100
Panel D: Optimal bandwidth (km)														
Inside	0.0008	0.0002	0.0290	−0.4042	−0.0207	−0.0211	0.0002	0.0001	−0.0005	0.0030	−0.0004	0.0012	0.0008	−0.0000
	(0.0036)	(0.0014)	(0.0232)	(4.9267)	(0.0260)	(0.0288)	(0.0011)	(0.0005)	(0.0008)	(0.0023)	(0.0005)	(0.0011)	(0.0010)	(0.0006)
Observations	1,175,095	1,320,551	500,061	549,861	844,202	752,374	1,454,386	2,399,172	1,311,801	363,877	1,508,651	459,522	945,760	1,100,704
Mean of dep. var.	0.4668	0.0689	0.2292	46.9729	0.5106	0.5816	0.0108	0.0090	0.0102	0.0150	0.0086	0.0123	0.0122	0.0101
Bandwidth	2.5692	2.8894	3.0525	3.3595	5.1565	5.3616	3.1809	5.2457	3.4649	4.6211	3.9845	5.8391	3.0770	3.5788
RD polynomial	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear
Airport × Segment FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster level	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport

Notes: The unit of observation is 1 km × 1 km grid cells. The no burning zones are areas within 15 km of an airport. This table checks potential sorting to validate the RD estimation around the no burning zone boundaries. *Inside* is an indicator for areas within the no burning zones. Columns (1)–(2) rule out the sorting of land usage. Columns (3)–(6) rule out the sorting of firms. Columns (7)–(8) rule out the sorting of farmers by excluding areas within 1 km of the no burning zone boundaries. Columns (9)–(12) rule out the sorting of farmers by looking at areas with differential ease of migration. Columns (13)–(14) rule out the sorting of airports by looking at preexisting airports. The optimal bandwidth in Panel D is selected using the procedure proposed by Imbens and Kalyanaraman (2012). Standard errors in parentheses are clustered at the airport level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

Table 10
The effect of no burning zones on agricultural fires: Robustness.

Dep. var. =	(1) Difference-in-Discontinuities analysis	(2)	(3)	(4)	(5) Alternative number of boundary segments	(6)	(7)	(8)	(9) % Cropland > 10%	(10)	(11) Major grain provinces	(12)
	#Fires	#Fires	#Fires > 0	#Fires > 0	#Fires	#Fires	#Fires > 0	#Fires > 0	#Fires	#Fires > 0	#Fires	#Fires > 0
Inside	0.0006		0.0003		0.0007	0.0004	−0.0001	−0.0001	0.0003	−0.0001	0.0014	−0.0002
	(0.0005)		(0.0004)		(0.0008)	(0.0007)	(0.0004)	(0.0004)	(0.0008)	(0.0005)	(0.0018)	(0.0009)
Inside × Post	0.0002	0.0006	−0.0004	0.0001								
	(0.0009)	(0.0008)	(0.0006)	(0.0005)								
RD polynomial	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear
RD polynomial × Post	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	No
Airport × Segment FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grid cell FE	No	Yes	No	Yes	No	No	No	No	No	No	No	No
Airport × Year FE	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	No
Number of segments	10	10	10	10	5	5	20	20	10	10	10	10
Cluster level	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport
Observations	1,110,368	1,110,368	1,110,368	1,110,368	1,021,851	1,089,873	1,778,652	1,876,313	970,182	1,648,696	492,724	674,229
Mean of dep. var.	0.0068	0.0068	0.0060	0.0060	0.0115	0.0114	0.0093	0.0092	0.0129	0.0105	0.0192	0.0153
Bandwidth (km)	4	4	4	4	2.2353	2.3849	3.8903	4.1023	2.4165	4.1061	2.5869	3.5339

Notes: The unit of observation is 1 km × 1 km grid cells. The no burning zones are areas within 15 km of an airport. This table checks the robustness of the RD estimation around the no burning zone boundaries. *Inside* is an indicator for areas within the no burning zones. Columns (1)–(4) use a difference-in-discontinuities specification utilizing the staggered opening of airports. Columns (5)–(8) use alternative number of boundary segments. Columns (9)–(10) use only airports around which the share of cropland accounts for more than 10%. Columns (11)–(12) focus on airports located in major grain-producing provinces. Standard errors in parentheses are clustered at the airport level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

the boundary. Most of the coefficients are small and statistically insignificant. There is a statistically significant jump in the number of firms and employees when using the 2 km bandwidth, but the economic magnitude is relatively small. Moreover, the positive coefficients would bias the results towards finding a significant effect of the regulation. In columns (7)–(8), I exclude areas within 1 km of the boundary,[30] since farmers living in such areas have strong incentive to cross the boundary for farming to avoid regulation. This type of sorting would also bias the results towards finding a significant effect of the regulation. I do not find substantial changes after excluding such areas. In columns (9)–(12), I split the sample into two groups by the ease of migration. I define migration as easy for areas within 1 km of major roads and railways and hard for other areas. I still cannot find a significant effect in these two types of areas. Finally, I test whether the results are driven by airports sorting into areas with few agricultural fires. To this end, I focus on airports established before 1999 in columns (13)–(14)and still cannot find a significant effect. In sum, these findings suggest that the results are unlikely to be driven by various forms of selective sorting.

5.4. Robustness

Tables 10 and 11 show the robustness of the baseline results. I report the results using the optimal bandwidth, except for a few specifications where fixed bandwidths are used due to the lack of a widely accepted optimal bandwidth algorithm. Overall, the difference in agricultural fires across the boundary is small and statistically insignificant across different samples and specifications.

Difference-in-discontinuities To fully utilize the panel structure of the data and control for confounding factors by exploiting within

[30] Appendix Table A23 shows that the results are similar if I exclude areas within 2 or 3 km of the boundary.

Table 11
The effect of no burning zones on agricultural fires: Robustness (Cont.)

Dep. var. =	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
	2-dimension RD		Triangular kernel		Quadratic polynomial		Geo. covariates		All covariates		Cluster: 2°× 2°		Conley SE	
	#Fires	#Fires > 0	#Fires	#Fires > 0	#Fires	#Fires > 0	#Fires	#Fires > 0	#Fires	#Fires > 0	#Fires	#Fires > 0	#Fires	#Fires > 0
Inside	0.0003	0.0003	0.0001	−0.0004	0.0006	0.0002	0.0003	−0.0001	−0.0006	−0.0001	0.0004	−0.0001	0.0004	−0.0001
	(0.0006)	(0.0004)	(0.0008)	(0.0004)	(0.0008)	(0.0005)	(0.0007)	(0.0004)	(0.0008)	(0.0004)	(0.0006)	(0.0004)	(0.0006)	(0.0003)
RD polynomial	Linear	Linear	Linear	Linear	Quadratic	Quadratic	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear
RD dimension	2	2	1	1	1	1	1	1	1	1	1	1	1	1
RD kernel	Uniform	Uniform	Triangle	Triangle	Uniform	Uniform	Uniform	Uniform	Uniform	Uniform	Uniform	Uniform	Uniform	Uniform
Geo. controls	No	No	No	No	No	No	Yes	Yes	Yes	Yes	No	No	No	No
All controls	No	No	No	No	No	No	No	No	Yes	Yes	No	No	No	No
Airport × segment FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster level	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	2°× 2°	2°× 2°	–	–
Observations	1,829,849	1,829,849	1,409,705	2,420,320	2,223,090	3,002,163	1,100,422	1,876,714	672,077	762,225	1,107,607	1,902,847	1,107,607	1,902,847
Mean of dep. var.	0.0110	0.0092	0.0112	0.0092	0.0109	0.0092	0.0114	0.0092	0.0080	0.0066	0.0114	0.0092	0.0114	0.0092
Bandwidth (km)	4	4	3.0822	5.2938	4.8627	6.5673	2.4063	4.1029	4.1038	4.6593	2.4227	4.1610	2.4227	4.1610

Notes: The unit of observation is 1 km × 1 km grid cells. The no burning zones are areas within 15 km of an airport. This table checks the robustness of the RD estimation around the no burning zone boundaries. *Inside* is an indicator for areas within the no burning zones. Columns (1)–(2) use a two-dimensional RD polynomial in latitude and longitude. Columns (3)–(4) use a triangular kernel, giving more weights to observations closer to the boundaries. Columns (5)–(6) use quadratic RD polynomial in distance to the boundaries. Columns (7)–(10) include the covariates used in balance checks. Columns (11)–(12) cluster the standard errors at the 2° × 2° level. Columns (13)–(14) adopt the Conley standard errors, assuming spatial correlation for observations within a 200 km radius and temporal correlation across 5 years. Standard errors in parentheses in columns (1)–(10) are clustered at the airport level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

grid cell variation, I use a difference-in-discontinuities specification following Grembi et al. (2016):

$$Y_{isjt} = \delta_{sj} + \mu_{jt} + \beta_1 Inside_{ij} + \beta_2 Distance_{ij} + \beta_3 Inside_{ij} \times Distance_{ij} + \beta_4 Inside_{ij} \times Post_{jt} + \beta_5 Distance_{ij} \times Post_{jt} + \beta_6 Inside_{ij} \times Distance_{ij} \times Post_{jt} + \epsilon_{isjt} \quad (8)$$

$$s.t. \quad -h \le Distance_{ij} \le h$$

where μ_{jt} is airport × year fixed effects, $Post_{jt}$ is a indicator equal to 1 if airport j is active at year t, and all other variables are as previously defined. I only include airports that opened between 2002 and 2015, in order to have at least one year of observations before the opening and at least one year of observations after the opening. The coefficient of interest is β_4, which measures the change in the discontinuities after the opening. I choose 4 km as the baseline bandwidth.[31] The results based on Eq. (8) are reported in columns (1) and (3) of Table 10. The coefficients on $Inside_{ij} \times Post_{jt}$ are still small and statistically insignificant. Columns (2) and (4) of Table 10 exploit within grid cell variation by further including grid cell fixed effects, and the results show no substantial change. Similar to difference-in-differences, this method requires that fires just inside and outside the boundary are on parallel trends before airport openings. I verify this assumption by estimating an event study variant of Eq. (8) and plot the results in Fig. 8. I do not find that fires around the boundary exhibit differential pre-trends.

Alternative FEs and samples Columns (5)–(8) of Table 10 use alternative number of boundary segments. In the baseline specification, I split each boundary into 10 segments to ensure comparison of observations in close proximity. Here I instead split each boundary into 5 or 20 segments to evaluate the robustness of the results. The results are not affected.

Columns (9)–(10) of Table 10 use only airports around which the share of cropland accounts for more than 10%. The insignificant difference in agricultural fires across the boundary might be driven by the lack of cropland around the airports. The results are still small and statistically insignificant. Columns (11)–(12) of Table 10 focus on airports located in major grain-producing provinces to check whether the insignificant results are driven by lower grain output, since agricultural fires come from burning crop residues.[32] The results barely change.

Two-dimensional RD Columns (1)–(2) of Table 11 use a two-dimensional linear RD polynomial in longitude and latitude (Dell, 2010). Although the treatment is purely determined by a one-dimensional variable that is the distance to the airport, such two-dimensional RD polynomial could flexibly control for geographic factors and provide checks on the baseline results. I choose 4 km as the baseline bandwidth and find that the coefficients are small and statistically insignificant.[33]

Weighting and quadratic polynomial Columns (3)–(4) of Table 11 use a triangular kernel, which gives higher weights to observations closer to the boundary.[34] The results are not affected. Columns (5)–(6) of Table 11 check robustness to quadratic polynomial in distance to the boundary.[35] The results are stable.

Including covariates Columns (7)–(10) of Table 11 add geographic covariates and other predetermined covariates used in the balance checks. In theory, adding such covariates should not affect the results but could improve the precision of the RD estimates (Imbens and Lemieux, 2008). The results stay small and statistically insignificant.

Spatial correlation Columns (11)–(12) of Table 11 cluster the standard errors at the 2° × 2° level. Cluster (13)–(14) of Table 11 report the results using the Conley standard errors, which allow for spatial correlation for observations within a 200 km radius and serial correlation across 5 years. The results are not affected.

5.5. Heterogeneity

To check whether the insignificant effects of no burning zones on agricultural fires represent general policy failures across regions and

[31] This is due to the lack of a widely accepted optimal bandwidth algorithm for the multi-dimensional RD design in the literature. In Appendix Table A24, I show that the results are robust to alternative bandwidths.

[32] The provinces are located in north, central, east, and northeast China, including Hubei, Hunan, Jiangxi, Jiangsu, Zhejiang, Hebei, Henan, Shandong, Shanxi, Heilongjiang, Jilin, and Liaoning.

[33] This is due to the lack of a widely accepted optimal bandwidth algorithm for the multi-dimensional RD design in the literature (Dell et al., 2018). In Appendix Table A25, I show that the results are robust to alternative bandwidths.

[34] The weight is $1 - abs(Distance_{ij})/h$, where $Distance_{ij}$ denotes the distance from each observation to the boundary and h is the bandwidth.

[35] I do not try orders higher than two since they are not recommended in RD designs and could generate undesirable results (Gelman and Imbens, 2019).

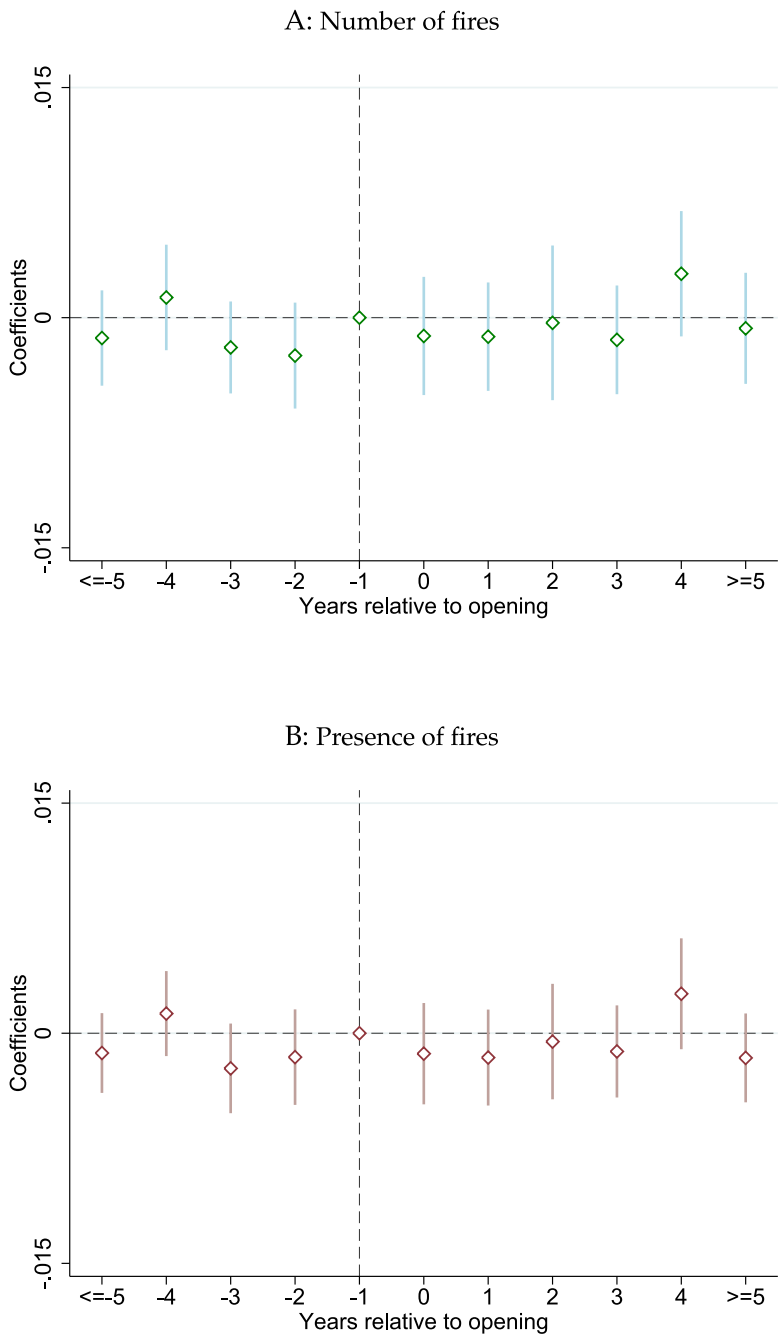


Fig. 8. No burning zones: Event study plot for difference-in-discontinuities estimation.
Notes: The unit of observation is 1 km × 1 km grid cells. The no burning zones are areas within 15 km of an airport. This figure conducts an event study plot for the difference-in-discontinuities estimation based on a 4 km bandwidth, utilizing the staggered opening of airports. *Presence of fires* is an indicator for observing at least one fire. The year before an airport opening is omitted as the reference group. The diamonds denote coefficients and bars denote the associated 95% confidence intervals, where standard errors used to construct the confidence intervals are clustered at the airport level.

years, I conduct RD estimation separately for each airport and for each year using the optimal bandwidth. The distribution of the coefficients is shown in Fig. 9. The airport-specific coefficients are small and centered at 0. Moreover, the coefficients are statistically insignificant for the vast majority of airports. For example, 190 out of 205 airports (≈ 93%) show insignificant coefficients in Panel A.[36] For the remaining 15 airports with significant coefficients, 4 airports show positive signs and 11 airports show negative signs. The year-specific coefficients also fluctuate around 0 and are generally insignificant. Overall, these results show that the weak effects of no burning zones on agricultural fires reflect a general policy failure.

I further test whether no burning zones work only at certain type of areas such as those subject to higher safety requirement or enjoying higher political status. Standard economic theory predicts that regulators would put more efforts into such areas, as the marginal benefits are higher. The results are reported in Table 12 using three fixed bandwidths and the optimal bandwidth. I first focus on upwind areas, which are defined as grid cells where the percentage of days upwind of the airport in a year is greater than the median percentage for that

[36] Results for Panel B are similar.

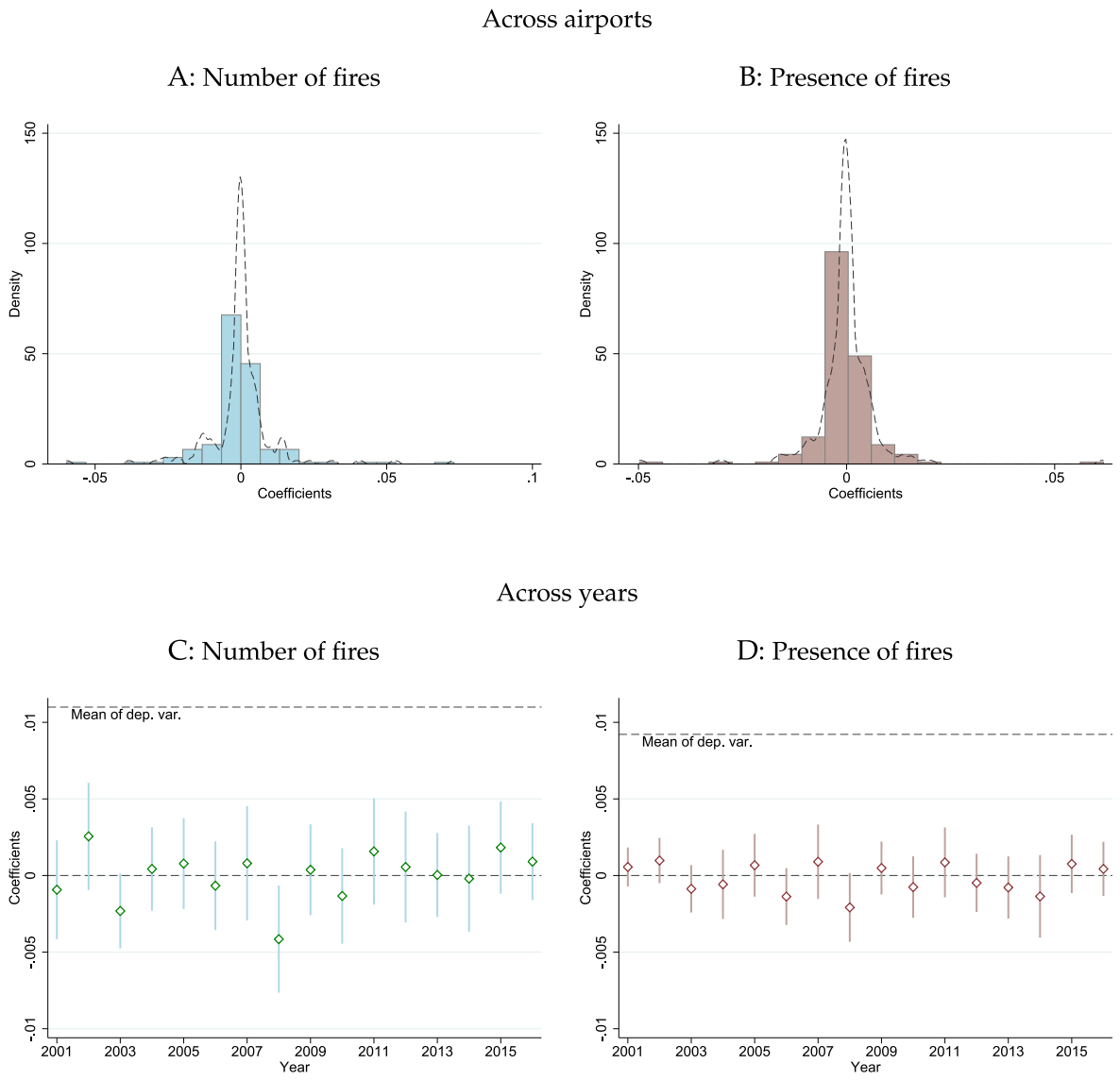


Fig. 9. No buring zones: Heterogeneity of RD estimates.
Notes: The unit of observation is 1 km × 1 km grid cells. The no burning zones are areas within 15 km of an airport. This figure conducts RD estimation separately for each airport and each year, using the optimal bandwidth calculated following the procedure proposed by Imbens and Kalyanaraman (2012), and shows little heterogeneity across airports or over years. *Presence of fires* is an indicator for observing at least one fire. Panel A and B show the distribution of airport-specific RD coefficients. Panel C and D show the year-specific RD coefficients (the diamonds) and the associated 95% confidence intervals (the bars), where standard errors used to construct the confidence intervals are clustered at the airport level.

airport and year.[37] Such areas are viewed as subject to higher safety requirement, as agricultural fires in such areas have a larger threat to the take-off and landing of the planes. If regulators selectively target such areas, then one should see a significant effect in such areas. The results in columns (1)–(2) and (7)–(8), however, show that the zones work neither in upwind areas nor in other areas. I next test whether the zones work in provincial cities.[38] These cities are typically more economically developed than other cities, and local leaders in these cities enjoy higher political status than other cities. Therefore, local leaders should have stronger incentive to fight agricultural fires for both economic and political reasons. However, the results in columns (3)–(4) and (9)–(10) show that the zones work neither in provincial cities nor in other cities. Finally, I test whether the zones work in cities where local leaders have higher promotion incentives. As before, I define higher promotion incentives as cities where the party secretaries are younger than 57.[39] The results in columns (5)–(6) and (11)–(12) show that the effects are still insignificant for both high- and low-incentive cities, except in column (11) where I use a much smaller bandwidth (2 km) and measure agricultural fires using a binary variable. For this coefficient that is statistically significant, the economic magnitude is relatively small compared to the sample mean. Further, the statistical significance disappears when using the optimal bandwidth. Hence, it is hard to conclude that the effectiveness of the zones changes with promotion incentives.

[37] The procedure to define upwind can be found in Section 3.2 and is also illustrated in Appendix Figure A1.

[38] These include 27 provincial capitals, 4 cities directly controlled by the central government, and 5 cities under separate state planning. The results stay unchanged if I only include provincial capitals.

[39] In Appendix Table A26, I measure promotion incentives using mayors' ages, as mayors may be responsible for economic activities. The results are similar.

Table 12
The effect of no burning zones on agricultural fires: Heterogeneity.

Dep. var. =	(1) Upwind area	(2) Non-upwind area	(3) Provincial city	(4) Non-prov. city	(5) High incentive	(6) Low incentive	(7) Upwind area	(8) Non-upwind area	(9) Provincial city	(10) Non-prov. city	(11) High incentive	(12) Low incentive
	Number of fires						Presence of fires					
						Panel A: Bandwidth = 2 km						
Inside	−0.0002	0.0011	0.0018	0.0003	−0.0000	0.0030	−0.0008	−0.0000	−0.0015	−0.0001	−0.0011**	0.0021
	(0.0011)	(0.0010)	(0.0017)	(0.0008)	(0.0008)	(0.0020)	(0.0006)	(0.0007)	(0.0011)	(0.0005)	(0.0005)	(0.0014)
Observations	427,125	455,768	131,710	783,192	629,566	141,339	427,125	455,768	131,710	783,192	629,566	141,339
Mean of dep. var.	0.0117	0.0111	0.0171	0.0106	0.0113	0.0193	0.0099	0.0092	0.0135	0.0090	0.0096	0.0159
						Panel B: Bandwidth = 4 km						
Inside	−0.0010	0.0012	0.0010	−0.0001	0.0002	0.0014	−0.0009	0.0005	−0.0004	−0.0001	−0.0004	0.0018
	(0.0010)	(0.0010)	(0.0022)	(0.0008)	(0.0008)	(0.0019)	(0.0006)	(0.0006)	(0.0012)	(0.0004)	(0.0005)	(0.0011)
Observations	854,726	910,774	261,819	1,568,030	1,258,846	281,844	854,726	910,774	261,819	1,568,030	1,258,846	281,844
Mean of dep. var.	0.0113	0.0105	0.0159	0.0102	0.0109	0.0180	0.0094	0.0088	0.0125	0.0087	0.0092	0.0148
						Panel C: Bandwidth = 6 km						
Inside	−0.0013	0.0005	−0.0001	−0.0004	−0.0001	−0.0004	−0.0008	0.0005	−0.0007	−0.0000	−0.0002	0.0012
	(0.0010)	(0.0011)	(0.0032)	(0.0007)	(0.0008)	(0.0020)	(0.0006)	(0.0006)	(0.0015)	(0.0004)	(0.0005)	(0.0011)
Observations	1,282,559	1,364,611	390,642	2,352,673	1,888,507	421,521	1,282,559	1,364,611	390,642	2,352,673	1,888,507	421,521
Mean of dep. var.	0.0110	0.0105	0.0155	0.0102	0.0109	0.0178	0.0093	0.0088	0.0124	0.0087	0.0092	0.0147
						Panel D: Bandwidth = Optimal bandwidth (km)						
Inside	−0.0009	0.0018	0.0023	0.0003	0.0007	0.0039*	−0.0007	0.0006	−0.0008	0.0001	−0.0001	0.0017
	(0.0010)	(0.0011)	(0.0017)	(0.0008)	(0.0009)	(0.0021)	(0.0006)	(0.0006)	(0.0012)	(0.0005)	(0.0005)	(0.0011)
Observations	881,969	557,823	215,318	902,363	993,730	197,732	953,146	972,217	266,888	2,014,149	1,571,582	262,851
Mean of dep. var.	0.0112	0.0110	0.0165	0.0104	0.0110	0.0192	0.0093	0.0088	0.0125	0.0086	0.0092	0.0149
Bandwidth	4.1242	2.4510	3.2760	2.3074	3.1583	2.8007	4.4602	4.2723	4.0744	5.1380	4.9945	3.7288
RD polynomial	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear	Linear
Airport × Segment FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster level	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport	Airport

Notes: The unit of observation is 1 km × 1 km grid cells. The no burning zones are areas within 15 km of an airport. This table checks the heterogeneity of the RD estimation around the no burning zone boundaries. *Presence of fires* is an indicator for observing at least one fire. *Inside* is an indicator for areas within the no burning zones. *Upwind area* denotes grid cells where the percentage of days upwind of the airport in a year is greater than the median percentage for that airport and year. *Provincial city* denotes provincial capitals and other cities which share roughly the same political status as provincial capitals. *High incentive* denotes that the city party secretary is younger than 57, as local leaders older than 57 have very low probability of being promoted. The optimal bandwidth in Panel D is selected using the procedure proposed by Imbens and Kalyanaraman (2012). Standard errors in parentheses are clustered at the airport level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

I also check separately for fires occurring in the daytime and nighttime. There is a concern that farmers may postpone burning that would otherwise occur in the daytime until nighttime to escape regulation. If this is true, one may not see an overall reduction in fires inside the no burning zones but could expect to see a reduction in the daytime; given that there are much more flights in the daytime, a possible reduction in fire in the daytime could still be meaningful. The results in Appendix Table A27, however, show that the zones work neither in the daytime nor nighttime.

6. Discussion

So far I have documented that the biomass power plants are effective in reducing burning while the no burning zones have little impact on burning. In this section I discuss a few remaining issues. First, one may be concerned about potential side effects of the plants. Appendix F.4 shows through a case study that a typical plant imposes little negative environmental externalities, such as air pollutants, water pollutants, and noise, on surrounding areas. Appendix H.1 shows that the plants can reduce ambient $PM_{2.5}$ concentrations by at least about 2%. Appendix H.2 shows that the plants do not deteriorate nearby water quality, which is consistent with the case study in Appendix F.4.

One may be interested in knowing whether areas falling into the overlapping catchment areas of more than one plant are different, which is, however, a priori unclear. The revenue from selling crop residues may not react to more than one potential buyer given the low and homogeneous price of electricity, which bounds the price of crop residues.[40] The explicit costs of collecting crop residues (e.g., labor and machine costs) may decrease with the number of potential buyers if more buyers lead to cheaper technologies and labor, although this may only happen when the number of buyers is sufficiently large. The explicit costs, which involve the gains from engaging in other activities, may increase if farmers work in the plants and earn higher wages in the presence of more than one plant, or if more plants lead to more prosperous local economic activities, although I empirically show that there is no change in employment and local economic activities. In sum, it is a priori unclear whether the economic benefits from selling crop residues would be different in the overlapping catchment areas. Consistent with such discussion, Appendix Table A19 demonstrates that these areas are no different from other areas.[41]

One may be concerned about differential crop types around the plants and airports, which may lead to the difference in crop residues and burning patterns. Although possible difference in crop types around the two types of areas does not affect the internal validity of my results, it may invalidate the contrast between these results. In Appendix I, I compare crop types around these two types of areas and do not find a substantial difference. Furthermore, the findings are robust to using a matching procedure that eliminates the difference in crop types across the plants and airports.

Throughout the analysis, grid cells are treated as if they are affected only by the plants or airports. However, some grid cells around the

[40] A plant may simply choose to not operate at its full capacity if the price of crop residues would be pushed up by more than one plant.

[41] Specifically, I include all plants and create an indicator *overlap*, which is equal to 1 if a grid cell around a certain plant also falls in the catchment areas of other plants that are active. I then fully interact this indicator with other terms in the baseline DID specification. The coefficients on the triple interaction terms are generally small and statistically insignificant, suggesting little difference between the overlapping areas and other areas.

plants (airports) may also be close to the airports (plants), which may raise the concern about compound treatment. To alleviate such concern, I redo the baseline analysis focusing on a sample in which the catchment areas of the plants do not overlap with those of the airports. Specifically, I drop all plants (airports) that are placed within 30 km of airports (plants). This trimmed sample includes 148 plants (78%) and 166 airports (81%). As shown in Appendix Table A21 and A29, the results are very similar to the baseline results.

Finally, one may also worry about the external validity of the spatial RD design around airports. Specifically, the no burning zones may be effective in curtailing fires at areas far away from the boundaries, which cannot be captured by the RD estimates. I therefore complement the RD design with a DID strategy that takes into consideration areas far from the boundaries in Appendix J. The results are small and statistically insignificant. Such results, together with the RD results, could be caused by a number of factors such as difficulty in patrolling the rural areas, lack of rule-abiding norms in rural areas, laziness, or incompetence of local officials. The current data do not allow me to disentangle these potential mechanisms. Instead, the goal of this paper is to provide credible evidence on the effects of different instruments on reducing agricultural fires.

7. Conclusion

In this paper, I test the effects of economic incentives and command and control on reducing agricultural fires using satellite-based high-resolution fire measures and two quasi-experiments: the opening of biomass power plants purchasing agricultural residues from nearby areas and the drawing of no burning zones banning agricultural fires. The results suggest that the plants could reduce fires by at least 30%. Such reduction is consistent with the plants increasing farmers' economic incentives to collect crop residues, as demonstrated not only by narrative evidence depicting farmers' reaction to the plants, but also by quantitative evidence showing smaller effects in areas that are more costly to collect crop residues. Using household survey data, I corroborate such mechanism by showing an increase in farmers' income after a plant opening.

Turning to the airports, I find no discernible difference in fires between areas just inside the no burning zones and areas just outside. Such finding is further complemented by a DID design that considers all areas inside the no burning zones, which extends the external validity of the baseline findings. By examining each airport and each year separately, I confirm that the results are not an artifact of potential heterogeneity across airports or years. Further heterogeneity analysis shows that the zones are dysfunctional even in areas with stringent regulatory enforcement, suggesting that the results represent a general policy failure.

A key contribution of this paper is that it examines the effectiveness of two common policy instruments in curbing agricultural fires rigorously using micro data, which has practical implications given that agricultural fires are still ubiquitous in the world. Particularly, in many developing countries plagued by agricultural fires, resorting to regulations or laws seems to be a straightforward and economical way to fight agricultural fires. The results in the China context show that, however, even in a country with strong state capacity and in areas with high safety requirements (i.e., areas around airports), a command and control policy could still be ineffective in curtailing agricultural fires. In contrast, the results show the success of instruments based on economic incentives. Note that the economic incentives created by the plants are slightly different from those created by traditional cash transfers, as the latter may incur considerable enforcement costs, which may undermine the policy effectiveness. In this vein, it would be interesting for future research to contrast the effectiveness of various types of economic incentive-based policy instruments in curbing agricultural fires and explore how the effectiveness interacts with enforcement costs.

CRediT authorship contribution statement

Yongwei Nian: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Data availability

The authors do not have permission to share data.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jdeveco.2023.103049.

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